

Follow the Workers: What LLM Research Agents Can and Cannot Do on a Weak Signal*

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Abstract

Language models are now asked to do research, not only to answer questions. In investment research the promise is risky: a model has read the history it is asked to predict, and it can state with confidence what it has never checked. We go beyond using a language model as a chat assistant and build a research pipeline of semi-autonomous Claude agents, each defined by a fixed prompt and kept away from the results by design: blinded to names and dates, scored by an evaluator they cannot see, and selected by a mechanical rule. We apply it to one question: do labor-market flows (job openings, hires, quits, layoffs, hours and wages, rebuilt as known on each date from 78 data vintages) predict which U.S. industries outperform? A preregistered test on 142 months says no at one month. A post-hoc iteration finds a slower, opposite effect consistent with the investment channel, promising but short of statistical evidence. The agents followed the protocol, better with a stronger model, yet found no signal that survived; sharing ideas between them had no detectable effect. Agents are disciplined research tools, not yet discoverers, and claims about how they behave need the same out-of-sample discipline as claims about alpha.

Keywords: industry rotation, labor flows, JOLTS, point-in-time data, LLM research agents, preregistration.

JEL codes: G11, G12, G14, C52, J63.

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1 Executive Summary

We tested whether public labor-market data can tell which U.S. industries will outperform. Each month, we score 13 industry groups on the year-over-year change in their hiring, quits, layoffs, hours and wages, as published at the time, and hold a market-neutral book long the top three and short the bottom three. A first, preregistered study showed that these flows say nothing about next month's returns. Its diagnosis pointed to a slower, opposite effect, which the data support: industries that hire and lose workers fastest underperform over the following year, as investment-based asset pricing predicts. Our post-hoc strategy, HQ12 (hires and quits, held twelve months), goes short these industries and long the slowest hirers. Its signal predicts returns a year ahead (IC +0.085, $t = 2.26$), and the book earns **a Sharpe ratio of +0.20 on data as known**, is positive in every window, passes six of seven robustness checks and stays profitable up to 50 bp of trading costs. The evidence is not yet strong enough to trade: the effect was found after the test period was opened, it is not significant once we account for the number of specifications tried, and its capacity is small. We conclude: **Do not implement** for now. HQ12 is in a paper forward test and should be judged on its return after a full year. Claude agents, used throughout, proved reliable auditors and disciplined researchers inside a strict protocol.

2 What We Tried: Research Design

2.1 Ideas: Motivation and Related Literature

The strategy. The labor market reports on an industry before its earnings do. Each month, the Bureau of Labor Statistics publishes job openings, hires, quits and layoffs by industry (JOLTS), along with hours and wages (CES). If this news reaches prices slowly (Hong et al., 2007), it should predict which industries outperform. Theory gives two channels of opposite sign. Under the demand channel (H1), rising hiring signals growing demand, and the industry outperforms within one to three months. Under the cost and investment channel (H2), heavy hiring raises adjustment costs, which lowers expected returns over one to two years (Belo et al., 2014). Wage growth fits neither cleanly: it squeezes margins, but labor risk raises expected returns (Kuehn et al., 2017; Donangelo et al., 2019). A third hypothesis (H3) is that labor-market tightness decides which channel wins. Only (H1) was preregistered; (H2) is tested post hoc. The benchmark to beat is industry momentum (Moskowitz and Grinblatt, 1999; Zarattini and Antonacci, 2024).

The AI research assistant. Language models now run research loops: they propose a hypothesis or a program, test it, and propose the next one. We borrow from three strands of this work and guard against three risks specific to finance. *Research loops.* Agents can search over programs and agent designs (Hu et al., 2025; Novikov et al., 2025; Lange et al., 2026; Zhang et al., 2026), including trading systems (Kvasiuk et al., 2026), trading desks of specialised agents (Xiao et al., 2024) and the mining of alpha factors (Wang et al., 2025; Li et al., 2025); ours propose signals in a restricted grammar and never see the evaluator. *Island models.* Parallel searches that exchange their best results can find more (Tanese, 1989; Romera-Paredes et al., 2024); we compare agents with and without sharing at equal budgets. *Self-critique.* Models can critique, debate or rewrite their own reasoning (Madaan et al., 2023; Shinn et al., 2023; Du et al., 2024; Spiess et al., 2025; Yin et al., 2025); we keep critique as a separate red-team step. *Look-ahead.* Language models can forecast returns from text (Lopez-Lira and Tang, 2023; Chen et al., 2023), but their forecasts can reflect what they learned about later events (Sarkar and Vafa, 2024); removing company names, as Glasserman and Lin (2023) do for news headlines, limits the damage. Our agents are blinded to names and dates, and our data are rebuilt as known on each date. *Biased judges.* LLM judges have systematic biases (Zheng et al., 2023), so ours is audited by hand. *Overfitting.* The more specifications are tried, the higher the bar a backtest must clear (Harvey et al., 2016; Harvey, 2017; Bailey and López de Prado, 2014); we preregister, open the test once and deflate every Sharpe ratio. One rule runs through all six: neither we nor the agents should be able to see the future.

Two questions. The paper answers two questions:

- Q1.** Do labor-market flows predict which U.S. industries outperform, net of costs and risk?
- Q2.** How should an LLM research assistant be integrated so that it helps without misleading us?

Sections 3.1 to 3.3 answer Q1, and Section 3.4 answers Q2.

2.2 Data: Labor Flows as They Were Known

Sources. All data are public and free. JOLTS gives monthly rates of job openings, hires, quits and layoffs for 13 industries, about two months after the fact. CES gives average weekly hours and hourly earnings of production workers, about one month after the fact. Labor-market tightness is $T = \log(V/U)$, total job openings over the number of unemployed, from FRED. Returns are the 49 value-weighted industry portfolios of the Ken French Data Library, with the five Fama–French factors and momentum. Series codes and links are in Appendix G.

Universe. We map the 49 industries to the 13 JOLTS groups and value-weight them within each group (Appendix H). The 13 groups are not 13 independent bets. Two groups share one CES series, and the participation ratio of the correlation matrix of their relative returns is 9.1. The mapping is also coarse: durable manufacturing, for instance, is 60% semiconductors by market capitalisation.

Data as they were known. Labor data are revised after their first release, and the revisions contain information that arrived later. A backtest on today’s data therefore looks ahead. We rebuilt each of the 78 labor series as it stood on every decision date, using ALFRED, the St. Louis Fed archive of past data vintages, in the spirit of real-time macroeconomic data sets (Croushore and Stark, 2001). The difference matters. Our best post-hoc strategy, HQ12[†], earns a test Sharpe ratio of +0.29 on revised data but +0.20 on data as known, and the two versions pick the same three top industries in only 39% of months. We therefore report results on data as known wherever both versions exist, flag the few figures available only on revised data, and Section 3.3 shows that our conclusions hold under first releases, revised data and fixed publication lags.

Sample. The research period, used to design signals and select models, runs from May 2004 to September 2014 (125 months). The test period runs from November 2014 to August 2026 (142 months) and was opened once. Figure 1 shows a consequence for H3: vacancies never outnumbered the unemployed in the research period, so a tightness effect could not be learned there.

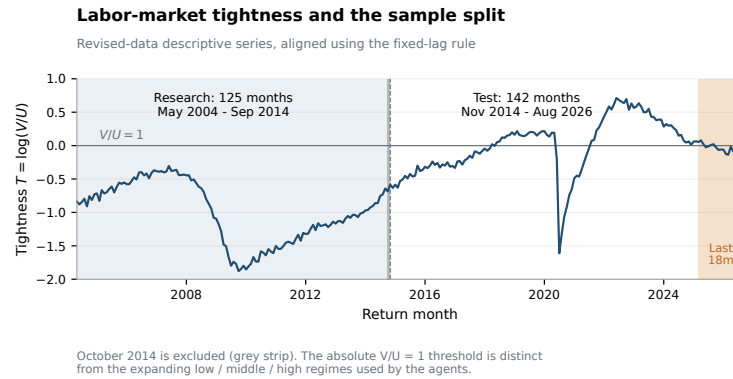


Figure 1: Labor-market tightness. Vacancies never exceeded the unemployed in the research period ($T < 0$ throughout), so a tightness effect learned there is extrapolated in the test.

2.3 Implementation: From Signals to Agents

From labor flows to signals. A signal asks a simple question: is this industry hiring (or quitting, or working longer hours) more than it was a year ago, compared with other industries? Take the hires rate in construction. We average it over the last three months available, which smooths out survey noise, and subtract the same average one year earlier:

$$F_{g,t} = \bar{x}_{g,t-\ell} - \bar{x}_{g,t-\ell-12}, \quad \bar{x}_{g,t} = \frac{1}{3} (x_{g,t} + x_{g,t-1} + x_{g,t-2}), \quad (1)$$

where $x_{g,t}$ is the series for industry group g in month t and ℓ is the publication delay (two months for JOLTS, one for CES). Comparing with the same months a year earlier removes seasonal patterns and each industry’s normal level. Each month, we then standardise the signal across the 13 industries, so that only the ranking matters: if every industry hires more, nothing changes.

Two studies. We ran two studies on the same data, with the same portfolio, the same rules and Claude in the loop (Table 1).

Study 1 was our first, preregistered pass. It tested the demand channel (H1): do labor flows predict next month’s industry returns? Its signals, models and pass/fail criteria were fixed before the test period was opened, and the test was run once. Its primary strategy, A3, adds to a ridge regression (Hoerl and Kennard, 1970) the signals proposed by the LLM agents described below. Study 1 did what a preregistered test is for: it gave a clean answer, no at one month (Section 3), and its diagnosis showed where to look next.

Study 2 builds on that diagnosis, as a quant researcher iterates on a signal. It makes three changes. It brings back wage growth, the only signal with a positive IC in both periods. It lets the research data set the signs that Study 1 had fixed in advance. And it asks the question the agents of Study 1 were never asked: what do labor flows predict a year ahead, rather than a month? At twelve months, hires and quits carry the sign the investment channel predicts (H2), and the resulting book, HQ12 (hires and quits, held twelve months), is the one we follow. Claude ran this iteration as an agent, with every stage frozen before it ran. Like any second iteration, Study 2 is partly fitted to what the first test revealed, so we mark its results with a dagger (†) and validate it the usual way, on new data, in a forward test.

Table 1: The two studies and their strategies.

Strategy	What it trades	Holding	Role of the LLM
<i>Study 1, preregistered: demand channel (H1), one-month signals</i>			
A0	Fixed signs on openings, hires, layoffs and hours	3 months	none
A1	Ridge regression on the five labor flows	3 months	none
A1-T	A1, with each flow interacted with tightness	3 months	none
A3 (primary)	A1 plus the independent agents’ features	3 months	features proposed by blinded agents
A4	A1 plus the communicating agents’ features	3 months	same, agents share reports
<i>Study 2, post hoc[†]: designed after a post-mortem, run by an iteration agent</i>			
W	Wage growth	3 months	post-mortem (P5); every stage frozen and run by the iteration agent (P6)
SC	All six signals, signs estimated on research data	3 months	
W+MOM	Wage growth and 12-month industry momentum	3 months	
HQ12	Short the industries that hire and quit fastest	12 months	

† Post hoc: decided after the test period was opened.

Portfolio and risk. Each month we rank the 13 industries on the signal, buy the top three and sell the bottom three, in equal weights. Each decision is held three months (twelve for HQ12): the book is the average of the last three (twelve) monthly portfolios, as in overlapping-portfolio momentum studies (Jegadeesh and Titman, 1993).

We hedge its market beta with a position in the market, then scale the whole book by a single number k_m :

$$k_m = \min \left\{ \frac{0.10}{\sqrt{12 x_m^\top \hat{\Sigma}_m x_m}}, \frac{2}{\sum_i |x_{m,i}|} \right\}, \quad w_m = k_m x_m, \quad (2)$$

where x_m is the unscaled book (the 13 industry weights and the market hedge) and $\hat{\Sigma}_m$ a Ledoit–Wolf covariance estimated over 60 months (Ledoit and Wolf, 2004). The first term targets a volatility of 10%, the second caps gross exposure at 2, and the book takes whichever is smaller.

Turnover and costs. Turnover is the share of the book traded each month, measured against the previous weights after they have drifted with returns, so that it counts only actual trades. We charge 10 bp per unit of turnover in the industry legs and 2 bp on the market hedge. HQ12 trades little, since only one twelve-month tranche is replaced each month: it turns its book over about 4.4 times a year on revised data, which costs about 0.4% a year, and it stays profitable up to 50 bp of trading costs. The monthly strategies of Study 1 trade about two and a half times as much. Capacity and the effect of the gross cap are discussed in Section 3.3.

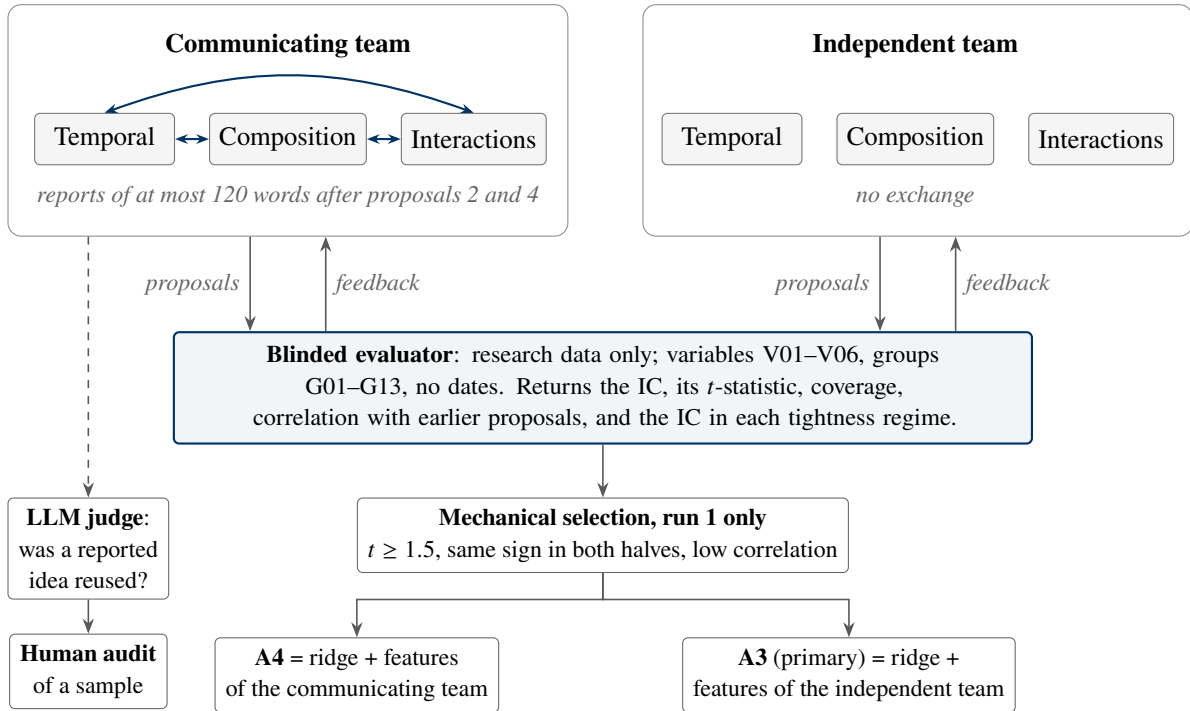


Figure 2: The agent experiment. Two teams of three agents with the same budget of six proposals each: one shares short reports, the other does not. Neither can see the evaluator, the names of the variables or the dates. The design is run three times; runs 2 and 3 only measure how much of the search is luck.

Where the agents come in. Figure 2 shows the agent experiment. Six autonomous agents propose new signals from the labor data. They are split into two teams of three, with the same budget of six proposals each. Each agent has one goal, stated in its prompt: find a formula that ranks the thirteen industries in the same order as their returns over the next month. The agent never sees the data or the returns. It sees only a short dictionary of the six labor variables under anonymous names (V01 to V06) with their units, the labels G01 to G13 for the industries, and no dates, so it cannot recall what happened to any real industry. A proposal is a short formula, much like a spreadsheet formula, built from a fixed menu of 14 operations such as changes, ratios, rankings and conditions on the state of the labor market (Appendix B). For example, $yoy(V05)$ is the change in variable V05 over the past year.

Each proposal goes to an evaluator, a program the agents can neither read nor modify. It computes the formula for every industry and month of the research period, and returns a short score. The main number is the information coefficient (IC): each month, the correlation between the ranking given by the formula and the ranking of next month's industry returns, averaged over the research period. The score also gives the IC's t -statistic, how often the signal was active, how close it is to earlier proposals, and its IC in each tightness regime. The agent's task is to raise this score, and it uses each score to decide what to try next. The two teams differ in one thing: after their second and fourth proposals, agents in the communicating team write a short report on what worked, and the others read it before proposing again. This is the island model of Section 2.1, and it asks whether sharing helps.

A fixed rule then selects, from the first of three runs, the features that are significant, stable over the research period and not redundant. The independent team's features enter the primary strategy (A3), the communicating team's enter A4. The second and third runs only measure how much of the search is luck. Separately, an LLM judge reads each report and the proposals that follow it to decide whether an idea was reused, and a person checks a sample of its decisions. We ran the experiment with Claude Sonnet (Study 1), and again with Claude Opus after the test had been opened (Study 1b).

In total, Claude played six roles, numbered in the order they occurred: P1, an audit of the data and code before any result existed; P2, the signal agents described above; P3, the judge; P4, a red team of the research results; P5, a post-mortem of Study 1; and P6, the iteration agent that rebuilt the analysis and ran Study 2 under a written rules file. Section 3.4 evaluates each.

How we kept the test honest

- **Same rules for both studies.** Every input as known on the decision date; every specification frozen and hashed before it ran; every result reported whatever its sign; Sharpe ratios deflated for every look taken (Bailey and López de Prado, 2014).
 - **Study 1 is confirmatory.** Its signals, models, primary strategy and five pass/fail gates were frozen before any test data were touched, and the test was opened once. We verified the freeze record against the repository.
 - **Study 2 is post hoc.** It was designed after the test was opened, so its test results cannot validate it. Its results carry a dagger (†), and only the forward test can confirm them.
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Prompts. Three prompts cover idea generation, coding support and robustness design. All are verbatim; outputs and our evaluation are in Section 3.4, transcripts and the rules that governed the iteration agent in Appendix E.

Idea generation: blinded signal agent (P2)

system prompt, opening

You are a quantitative researcher. You will propose one feature per turn to predict which of 13 anonymous groups (G01 to G13) will have higher returns next month relative to the others. You see only the variable dictionary below, the allowed operations, and summary statistics of your previous submissions. You do not know what the groups, variables or time periods are. Do not guess them and do not use knowledge about any real industry, company or date. [variable dictionary, grammar and JSON format follow]

Coding support: data and code audit (P1)

user prompt

Find mapping errors, timestamp or look-ahead errors, and code bugs in the material below. For each problem, give the exact line or row, why it is wrong, and a test that would confirm it. Do not report style issues. [followed by the series manifest, the industry crosswalk and the as-of code]

Robustness design: red team (P4)

user prompt

Give 10 distinct reasons these results could be spurious, ranked by likelihood. For each, name a test that can be run with this data and the result that would make us say "do not implement". [followed by the research-period result tables, without dates or names]

3 What We Learned: Results

Summary of findings

- **H1, demand channel: rejected.** The preregistered one-month strategies did not predict industry returns; the primary strategy's forecasts did no better than a placebo ($p = 0.72$).
- **H2, slow cost channel: promising, not established[†].** Shorting the industries that hire and quit fastest for twelve months (HQ12) earns a test Sharpe ratio of +0.20 on data as known and is positive in every window. It was designed after the test was opened.
- **H3, tightness: untestable as designed.** Vacancies never exceeded the unemployed in the research period.
- **Agents: discipline without discovery.** They followed the protocol, better with a stronger model, but none of their signals survived the test, and our explanations of their failures did not hold up when tested.

Table 2: Net Sharpe ratios on data as known: the one-time test and the windows the brief asks for.

	Research	Test	Full	Post-2010	Last 18m
<i>Study 2, post hoc[†]</i>					
HQ12 hires and quits, 12 months	+0.54	+0.20	+0.31	+0.23	+0.14
W wage growth	+0.11	+0.13	+0.13	+0.10	−1.31
W+MOM wage growth and momentum	+0.25	+0.46	+0.37	+0.31	−0.94
SC signed composite	+0.34	+0.24	+0.28	+0.12	+1.38
<i>Study 1, preregistered: the one-time test</i>					
A3 agent features (primary)	−0.13	−0.58	−0.44	−0.62	+0.85
A4 communicating agents	−0.18	−0.79	−0.56	−0.74	+1.23

Annualised, net of 10 bp industry and 2 bp hedge costs, labor data as published on each decision date. Test: November 2014 to August 2026 (142 months); last 18 months: March 2025 to August 2026. For Study 1, the research column uses the 88 months in which every model had a forecast, and the full sample starts in June 2007. On revised data, HQ12's test Sharpe ratio is +0.29. A0, A1 and A1-T are in Appendix B. [†] Post hoc: decided after the test period was opened.

3.1 Style and Risk Exposures: What Else the Books Hold

We regress each book's net return on the five Fama–French factors (Fama and French, 2015), momentum (Carhart, 1997) and industry momentum (full loadings in Appendix C). HQ12 is short profitability (RMW −0.36, $t = -4.5$, revised data): shorting the industries that hire and quit fastest means shorting profitable, growing industries. It does not load on the investment factor (CMA), so it is not the known investment premium in disguise, and its alpha is positive but not significant (+0.16% a month on data as known, $t = 0.92$). Study 1's primary strategy leaned the other way, long profitability (RMW +0.19). **For Q1**, what labor flows capture beyond profitability is positive but too small to measure with confidence.

3.2 Performance over Time: Positive Everywhere, Not Every Year

HQ12 is positive in every window of Table 2: +0.31 over the full sample, +0.23 after 2010 and +0.14 over the last 18 months. So is the signed composite, but its signs change with the data version, so we read little into it. These windows hide an uneven path, however. Within the test, HQ12 lost money in its first half (2014 to 2020, −0.23) and more than made it back in the second (+0.49); the longer windows are positive because the later gains outweigh that loss. Wage growth worked until 2024 and lost heavily over the last 18 months (−1.31). Study 1 lost over the full sample and after 2010, and its gain over the last 18 months rests on an IC that is statistically zero ($t = 0.44$). Industry momentum, the benchmark to beat, earned +0.03 over the test. **For Q1**, the slow hiring signal is the only stable one across windows, and only half of the test supports it.

3.3 Robustness and Limits: Why the Demand Channel Failed, What Might Survive

Why the demand channel failed. Study 1's answer holds. Its primary strategy stays negative under every data version, cost level, dropped industry, gross cap and weighting scheme, and in every sub-period except the last 18 months; its 90% block-bootstrap interval (Künsch, 1989), $[-0.99, -0.16]$, excludes zero; its forecasts did no better than a placebo; and no preregistered strategy beat another once corrected for multiple comparisons (Holm, 1979). The diagnosis has four parts, each addressed by Study 2. Layoffs and hours carried the opposite sign to the one imposed. Hiring predicts nothing at one month, but lower returns at twelve. The slack regime that H3 relied on occurred in 3 of 142 test months. And the two agent features of the primary strategy were active only in those months. The warning signs were already visible on research data, where every Study 1 strategy had a negative Sharpe ratio and cross-validation chose the strongest penalty for every model.

What might survive: a slow cost channel[†]. HQ12's sign is the one the investment channel predicts (Belo et al., 2014). Its twelve-month horizon, however, was chosen from a profile that included the test: on research data alone, its twelve-month IC is not significant, and that sample would not have selected it. Figure 3 shows the profile: nothing at one month, a positive IC from nine to 24 months. On data as known, HQ12 passes six of the seven checks declared before its battery ran, and fails one: the first half of the test lost money (Table 3). Its alpha and deflated Sharpe ratio remain far from the thresholds declared with the variants. The other variants are weaker: W has no alpha, W+MOM draws its return from momentum, and SC's signs depend on the data version.

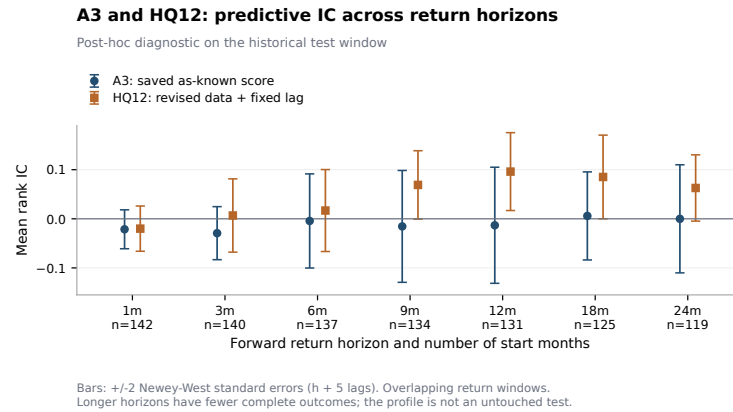


Figure 3: Where each score predicts. The primary score of Study 1 has no IC at any horizon; the HQ12 score has none at one month and a positive IC from nine months on. Revised data.[†]

Table 3: HQ12 robustness battery, data as known, test window unless stated.

Check (declared before the battery ran)	Result	Met
Each component alone, held 12 months (Sharpe): hires; quits	+0.19; +0.17	yes
12-month IC (t); 1-month IC t	+0.085 (2.26); -0.42	yes
Holding 9 and 18 months (Sharpe)	+0.16; +0.11	yes
Both test halves positive	-0.23; +0.49	no
Break-even trading cost above 25 bp	50 bp	yes
Drop one industry: lowest and highest Sharpe	+0.05; +0.36	yes
Bootstrap 90% interval excludes zero: test; full sample	[-0.14, +0.57]; [+0.02, +0.60]	yes (full)
<i>From the reading rule declared with the variants</i>		
Factor alpha t (test; full sample)	0.92; 1.81	no
Deflated Sharpe ratio (162 looks)	0.02	no

Newey–West standard errors (Newey and West, 1987): 12 lags for the alpha, $h + 5$ for ICs at horizon h . † Post hoc: decided after the test period was opened.

Limits. Even a genuine signal could not have earned much here. The fundamental law of active management (Grinold, 1989; Clarke et al., 2002) links the information ratio to the quality of each bet (IC), the number of independent bets per year (BR) and the share of the signal that reaches the portfolio (TC):

$$IR \approx IC \times \sqrt{BR} \times TC. \quad (3)$$

The 13 industries behave like nine independent bets, so a monthly book has $BR \approx 9 \times 12 = 108$, and even a good IC of 0.05 caps the ratio near 0.5. HQ12 renews its bets once a year across about ten effective industries ($BR \approx 10$) and passes about half of its signal to the portfolio ($TC \approx 0.48$), so the law implies $0.085 \times \sqrt{10} \times 0.48 \approx 0.13$. That is the same order as the +0.20 it realised and well within sampling error, since a Sharpe ratio measured over 142 months has a standard error of about 0.3: the result is not too good to be true. Capacity is small (about \$1.2 million before trading reaches 1% of the least liquid ETF’s daily volume), and the gross cap binds in most months without hurting the Sharpe ratio. H3 could not be tested, HQ12’s horizon was chosen with the test in view, the agents searched only at one month, and with 142 months only a Sharpe ratio near 0.6 reaches $t \approx 2$. The forward test frozen on 3 October 2026 (Appendix D) holds one tranche for a quarter, where HQ12 shows no IC; only that tranche’s twelve-month return will be informative.

Answer to Q1. Labor flows do not predict which industries outperform over the next month. Over the next year, the data are consistent with the investment channel, but the evidence is post hoc and short of significance.

3.4 Evaluating the AI Research Assistant: Discipline without Discovery

Claude took every AI role: Sonnet 5.5 for the 199 calls of Study 1, Opus 5.5 for the 294 calls of Study 1b, and later Claude models for the post-mortem and the iteration. Each interaction is documented, with transcripts and our critiques, in Appendix E. Its six roles, numbered P1 to P6 in the order they occurred, are defined in Section 2.3; we evaluate Claude as an assistant, as a researcher and as a judge.

Claude as assistant: a useful auditor that moves the goalposts. The audit of the data and code (P1) caught every real gap in the mapping, but five of its 25 findings were claims the files did not support, and it presented design choices as “confirmed errors”. The red team (P4) read the research tables correctly, yet asserted a p -value it had never computed and proposed new pass/fail thresholds after seeing the results. Both errors are typical of language models: they can state with confidence what they have not checked, a failure known as hallucination (Ji et al., 2023), and specialised quantitative work leaves little room for it. The post-mortem (P5) found the right cause of Study 1’s failure, but its fixes were read off the test results, and the wage signal it called stable lost heavily over the last 18 months. We checked every finding, kept the registered tests and labelled every later change post hoc.

The iteration agent (P6) is the counter-example, and the clearest gain. It rebuilt the analysis, reproduced Study 1’s returns to 10^{-16} , ran every stage of Study 2 with each number tied to the file it comes from, and froze each stage before running it, work that would have taken us far longer by hand. When it wrongly assumed that August JOLTS data would not be out by 29 September, its own pre-committed check stopped the run. **The difference lies in the setting, not the model:** the red team and the post-mortem could change the rules, the iteration agent could not, because its rules were enforced by code rather than stated in a prompt.

Claude as researcher: protocol followed, nothing found. The agents followed the blind protocol, and a stronger model followed it better (Table 4): no call tried to recover the hidden names or dates, and Opus made more valid proposals than Sonnet. Neither found a signal that survived. A feature’s research IC barely predicts its test IC (Figure 4), and none of the eight selected features met the test criterion, though two of Opus’s kept a small positive test IC. **A stronger model improved discipline, not discovery.**

Table 4: The agents in numbers: a stronger model followed the protocol better and found no more signal.

	Study 1 (Sonnet 5.5)	Study 1b (Opus 5.5) [†]
Valid proposals	98 / 108	106 / 108
Reports admitted (communicating team)	7 / 18	15 / 18
Usable judge answers	8 / 36	84 / 88
Correlation of research and test IC	0.22	0.37
Selected features meeting the test criterion	0 / 4	0 / 4

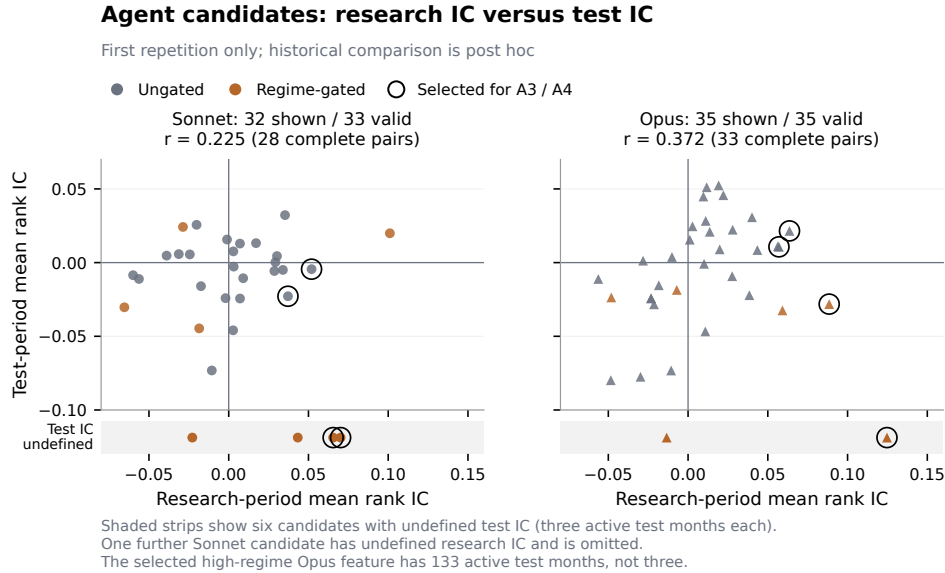


Figure 4: Research IC of agent features barely predicts their test IC. The selected regime-gated features were active in 3 test months.[†]

Islands: no detectable gain from sharing. The island model is the most distinctive part of our design (Section 2.1). **In our setting, sharing had no detectable effect.** In both studies, the gap in best research IC between the communicating and the independent team lies within the spread across the three runs (permutation $p = 0.30$ for Sonnet and 0.10 for Opus; Appendix C). Two details matter. With Sonnet, sharing barely happened: 11 of 18 reports were rejected, mostly for running over 120 words. With Opus, which delivered 15 of 18 reports, the

communicating team did better in all three runs, the most extreme ordering three runs allow. Sharing may therefore help a model that follows the exchange protocol, but three runs cannot tell. Our setting was also unfavourable to islands: they pay off when parallel searches tackle different problems and have something worth passing on, whereas our two teams searched the same problem for a signal that may not exist.

Claude as judge, and testing our own explanations. To measure whether ideas spread, an LLM judge read each report and the proposals that followed it (P3). It overstated uptake: on the 17 Opus pairs we checked by hand, it said an idea had been reused 10 times and the human 3 times, and all seven disagreements were false yeses. It matched vocabulary rather than content.

Our critiques proposed two explanations, each with a fix, and we tested both. If the judge matches vocabulary, requiring it to quote the exact claim it relies on should help: it fixed the format of Sonnet’s answers (8 of 8 usable, against 2 before) but not Opus’s judgment, which still said yes falsely 8 times in 17. If the agents built regime gates because their feedback showed the IC in each tightness regime, removing that split should remove the gates: it did not (4 of 48 valid proposals with the split, 3 of 50 without). The rerun produced fewer gates than the original study, so this test is weak, but it gives no support to our explanation. The selection data point to another one: gated features were measured on 43 to 48 consecutive research months, against 120 or more for the others, and they were overrepresented among the selected features (2 of 4 in each study, against 9 of 33 and 6 of 35 among the candidates). The selection rule, not the feedback, may have rewarded them. **Our first explanations of the agents’ behaviour were plausible, and they did not survive a test.**

Next time, we would require a minimum of active months for selection, check the judge’s quotes by text search, search at several horizons and run more repetitions.

Answer to Q2. An LLM assistant helped wherever the protocol kept it away from the results, and every claim about its behaviour, including our own, had to be tested.

4 Conclusion

We asked two questions.

Q1: do labor-market flows predict which U.S. industries outperform, net of costs and risk? **Not over the next month.** Study 1, preregistered and tested once on 142 months, found no predictive power, and that answer holds under every check. Over the next year, the data are consistent with the investment channel: the industries that hire and lose workers fastest earned lower returns, and HQ12, which shorts them, earned a Sharpe ratio of +0.20 on data as known. But this was found after the test was opened, and it does not survive a correction for the number of looks taken. We conclude: **do not implement**, and follow HQ12 out of sample, at its own twelve-month horizon.

Q2: how should an LLM research assistant be integrated so that it helps without misleading us? **By giving it the parts of research it does well and keeping it away from the results.** Claude was a fast, wide auditor and a disciplined signal searcher, and a stronger model was more disciplined still. But it found nothing that survived the test, it moved the goalposts whenever it saw results, and it was an unreliable judge of its own agents. What made it useful was the protocol around it: blinding, an evaluator out of its reach, frozen specifications and human checks. And when we tried to explain its failures, our own explanations did not survive testing either. Claims about how an AI behaves need the same out-of-sample discipline as claims about alpha.

Three extensions follow. Stock-level baskets built from NAICS codes would give the signal far more independent bets than thirteen industries. Agents asked to search at several horizons could test what Study 1’s agents were never asked. And the HQ12 tranche bought on 30 September 2026 should be evaluated after its full twelve months, not after one quarter.

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Appendix A. Names used in this report

Repository. Code, data-building scripts, frozen specifications, run logs and every table and figure of this report: <https://github.com/pieropls/MFE230GA>.

The repository and the result files were renamed on 4 October 2026. Result files frozen before that date keep the old codes inside, so they stay byte-identical.

Table 5: Name mapping.

Old name	Name in this report
frozen, sealed or Sonnet study	Study 1: preregistered (Alex), folder study1_preregistered/
Opus follow-up	Study 1b: exploratory follow-up (Opus)
v2, v3, v4 runs	Study 2: post-hoc iteration; 2.1 variants (was v2), 2.2 robustness of HQ12 (was v3), 2.3 additional robustness (was v4)
A0, A1, A1-T, A3, A4	A0 fixed rule; A1 ridge; A1-T ridge $\times$ tightness; A3 agent features, independent (primary); A4 agent features, communicating
V2a, V2b, V2c, V2d	W wage growth; SC signed composite; W+MOM wage growth and momentum; HQ12 hires and quits, held twelve months
P1 to P6	P1 data and code audit; P2 signal agents; P3 judge; P4 red team; P5 post-mortem review; P6 iteration agent
islands, sealed test, F1–F5	communicating team; one-time test; openings, hires, quits, layoffs, hours

Appendix B. Study 1 in full: preregistered (Alex)

Everything in this section labelled confirmatory is read from Study 1’s saved files and is unchanged: `results.json` and the ledgers next to it, under `study1_preregistered/outputs/follow_the_workers/outputs/sealed/initial/`. The diagnostics labelled post hoc were computed afterwards from the same saved files and public data.

Design

Thirteen industry groups (Appendix H); signals F1–F5 as known at each decision date; five strategies (Table 1); research May 2004 to September 2014, test November 2014 to August 2026; top three against bottom three, three monthly tranches, beta hedge, 10% volatility target, gross cap 2, 10 bp and 2 bp costs. Five gates decide the verdict for the primary: G1 timing (pass), G2 prediction (IC $t \geq 2$; fail), G3 incremental alpha ($t \geq 2$; fail), G4 robustness (positive IC in both halves, placebo $p < 0.05$, no industry above half the profit; fail), G5 implementable (Sharpe > 0 at 25 bp, capacity documented; fail). The freeze record was checked against the files (Table 10). The score-to-weight mapping, with $s_{m,g}$ the score of group g known at the end of month $m - 1$, is below; Equation (2) in the main text is its scaling step.

$$\tau_{m,g} = \begin{cases} +\frac{1}{3} & g \in \text{top 3 of } s_{m,\cdot} \\ -\frac{1}{3} & g \in \text{bottom 3 of } s_{m,\cdot} \\ 0 & \text{otherwise} \end{cases} \quad b_m = \frac{1}{H} \sum_{j=0}^{H-1} \tau_{m-j}, \quad H = 3, \quad (4)$$

$$h_m = - \sum_g b_{m,g} \hat{\beta}_{m,g}, \quad x_m = (b_{m,1}, \dots, b_{m,13}, h_m), \quad (5)$$

$$k_m = \min \left(\frac{0.10}{\sqrt{12 x_m^\top \hat{\Sigma}_m x_m}}, \frac{2}{\|x_m\|_1} \right), \quad w_m = k_m x_m, \quad (6)$$

$$r_m^{\text{net}} = \sum_g w_{m,g} r_{m,g}^e + w_{m,M} \text{MKT}_m - c \sum_g |w_{m,g} - \tilde{w}_{m,g}| - c_h |w_{m,M} - \tilde{w}_{m,M}|. \quad (7)$$

where $s_{m,g}$ is the score of group g known at the end of month $m - 1$, $\hat{\beta}_{m,g}$ and the Ledoit–Wolf covariance $\hat{\Sigma}_m$ (13 groups plus the market) use the 60 months ending in $m - 2$ (at least 36), $r_{m,g}^e$ is the group’s excess return, $\tilde{w}$ are

the previous weights after drift, $c = 10$ bp and $c_h = 2$ bp.

Table 6: Study 1: all five strategies in the test window.

confirmatory

strategy	Sharpe	Net p.a.	Vol.	Max DD	Turnover	Costs p.a.
A0 fixed rule	−0.36	−3.5%	9.7%	−40.3%	0.83	1.01%
A1 ridge	−0.51	−4.0%	7.9%	−37.7%	0.91	1.11%
A3 primary	−0.58	−4.6%	7.8%	−41.1%	0.91	1.11%
A4 communicating agents	−0.79	−6.1%	7.8%	−53.3%	1.11	1.35%
Industry momentum 12-1	+0.03	+0.3%	9.4%	−20.8%	0.53	0.65%

Table 7: Study 1 robustness, full table: the four data versions, costs, sub-periods, drop-one-group, gates, and the diagnostics added in Study 2.3.

confirmatory / post hoc (last four rows)

Check (test window, as-known data unless stated)	A3 agent features (primary)	A0 fixed rule
Test Sharpe, data as known	−0.58	−0.36
Test Sharpe, data first release	−0.52	−0.16
Test Sharpe, data revised	−0.56	−0.26
Test Sharpe, data fixed lag	−0.53	−0.27
Net return p.a.	−4.6%	−3.5%
IC, one month (t)	−0.021 (−1.08)	−0.020 (−0.71)
Factor alpha per month (t)	−0.38% (−1.86)	−0.25% (−0.92)
Placebo p (circular shift)	0.72	n/a (primary only)
Holm-adjusted p, IC and return	1.00, 1.00 (A3 vs A1)	1.00, 1.00 (A1 vs A0)
Deflated Sharpe, research (41 trials)	0.003	0.000
Sharpe at 0 / 10 / 25 bp	−0.45 / −0.58 / −0.79	n/a / −0.36 / n/a
Sharpe, first test half	−0.72	−0.55
Sharpe, second test half	−0.49	−0.20
Sharpe, excluding Mar-Dec 2020	−0.51	−0.18
Sharpe, last 18 months	+0.85	−1.56
Drop one group, Sharpe range	−0.70 (Wholesale) to −0.24 (Durable mfg)	n/a (primary only)
Gates G1 timing / G2 prediction / G3 alpha / G4 robustness / G5 implementable	pass / fail / fail / fail / fail	n/a (primary only)
Verdict	Do not implement	not primary
Block-bootstrap 90% interval, test Sharpe	[−0.99, −0.16]	[−0.91, +0.20]
A3 score IC at 1 / 6 / 12 / 24 months (t)	−0.021 (−1.08) / −0.004 (−0.09) / −0.013 (−0.22) / −0.000 (−0.00)	n/a
A3 Sharpe with cap 2 / cap 3 / no cap	−0.58 / −0.61 / −0.62	n/a
A3 Sharpe, rank-weighted book	−0.49	n/a

Table 8: Risk overlay of every Study 1 and Study 1b strategy: how often the cap bound, ex-ante and realised volatility.

confirmatory

Study, strategy	Cap binds	Ex-ante vol	Realised vol	Gross leverage	Hedge
Study 1 A0	80%	8.0%	9.7%	1.96	0.15
Study 1 A1	88%	7.5%	7.9%	1.98	0.13
Study 1 A1-T	82%	7.5%	8.2%	1.97	0.12
Study 1 A3	87%	7.5%	7.8%	1.98	0.13
Study 1 A4	91%	7.3%	7.8%	1.99	0.13
Study 1b A3	92%	7.3%	7.5%	1.99	0.15
Study 1b A4	93%	7.3%	8.4%	1.99	0.11

Table 9: Factor loadings of the Study 1 books (FF5, momentum, industry momentum).

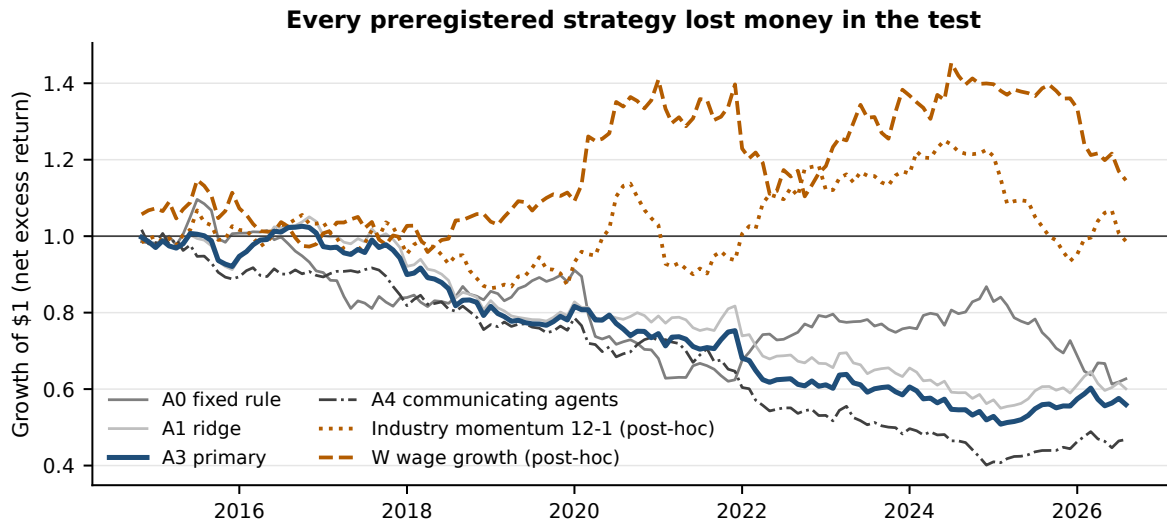
confirmatory

Factor	A0 fixed rule	A1 ridge	A3 primary
Alpha (per month)	−0.25% (−0.92)	−0.35% (−1.72)	−0.38% (−1.86)
Market	−0.01 (−0.31)	−0.01 (−0.20)	−0.03 (−0.43)
SMB	+0.02 (+0.21)	+0.12 (+1.01)	+0.10 (+0.89)
HML	+0.09 (+0.60)	−0.15 (−1.97)	−0.15 (−2.01)
RMW	+0.00 (+0.05)	+0.21 (+2.33)	+0.19 (+2.16)
CMA	−0.04 (−0.21)	+0.05 (+0.47)	+0.03 (+0.26)
Momentum	−0.06 (−0.72)	+0.01 (+0.15)	−0.01 (−0.17)
Industry momentum	+0.22 (+1.93)	−0.02 (−0.21)	−0.01 (−0.12)

Table 10: Freeze record, checked against the files in the repository.

confirmatory

check	ok	value
SHA-256 of FREEZE.json equals evaluation_started.freeze_sha256	True	60ea05294ffc856e
Freeze recorded (UTC)	True	2026-10-02T06:00:09.537758+00:00
Test opened (UTC)	True	2026-10-02T06:00:09.578162+00:00
Test opened after the freeze	True	0.040 s later
Files in the freeze inventory	True	525
of which present in the repository	True	338
present files whose hash matches the inventory	True	337
present files that differ	True	outputs/human_migration_review.md
First test decision in the freeze record	True	2014-10-31

**Figure 5:** Cumulative net returns in the test: every Study 1 strategy lost money; the post-hoc wage-growth book gained until 2024 and gave much of it back. Revised data for W.

confirmatory (A0–A4) / post hoc (momentum, W)

Diagnostics of the failure

Table 11: Tightness regimes: months per bin under the expanding-tercile rule and under a 12-month change rule. *post hoc*

Months in regime	Research 2004–14	Test 2014–26
months	125	142
expanding low	49	3
expanding middle	26	6
expanding high	50	133
T rising (12m)	93	88
T falling (12m)	32	54
V/U > 1	0	75
T min	−1.88	−1.61
T max	−0.30	+0.71

Table 12: Signal IC when tightness rose or fell over the previous 12 months.

post hoc

Signal	T rising (182 m)	T falling (86 m)	Difference
F1 openings	+0.001 (+0.05)	−0.019 (−0.47)	+0.019
F2 hires	+0.032 (+1.51)	−0.026 (−0.74)	+0.059
F3 quits	+0.024 (+1.32)	−0.024 (−0.76)	+0.048
F4 layoffs	+0.013 (+0.53)	+0.060 (+1.69)	−0.046
F5 hours	−0.002 (−0.13)	−0.026 (−1.00)	+0.024
W wage growth	+0.033 (+1.67)	+0.053 (+2.04)	−0.020
A0 composite (prereg.)	+0.003 (+0.13)	−0.045 (−1.28)	+0.048
Industry momentum 12-1	+0.036 (+1.68)	−0.014 (−0.33)	+0.050

Table 13: A3 profit and loss by industry in the test.

post hoc

component	Contribution (pp)	Months long	Months short
Mining & logging	+7.2	33%	36%
Construction	−19.2	36%	39%
Durable mfg	+1.8	31%	24%
Nondurable mfg	+5.4	32%	35%
Wholesale	+14.9	31%	42%
Retail	+0.2	39%	35%
Transport & utilities	+1.4	47%	30%
Information	−5.8	30%	37%
Finance & insurance	−16.3	34%	45%
Real estate	−1.6	24%	45%
Prof. & business svcs	−12.2	18%	47%
Health care	−26.2	41%	36%
Accommodation & food	+4.5	46%	22%
Market hedge	+4.6		
Trading costs	−13.1		

Table 14: Horizon profile of the A3 score against the HQ12 score, test window: rank IC (t in brackets), revised data; the numbers behind [Figure 3](#). *post hoc*

Horizon	A3 agent features (Study 1 primary)	HQ12 (Study 2)
1 month	-0.021 (-1.08)	-0.020 (-0.87)
3 months	-0.029 (-1.08)	+0.007 (+0.18)
6 months	-0.004 (-0.09)	+0.017 (+0.40)
9 months	-0.015 (-0.27)	+0.069 (+1.98)
12 months	-0.013 (-0.22)	+0.096 (+2.43)
18 months	+0.006 (+0.13)	+0.085 (+2.00)
24 months	-0.000 (-0.00)	+0.063 (+1.86)

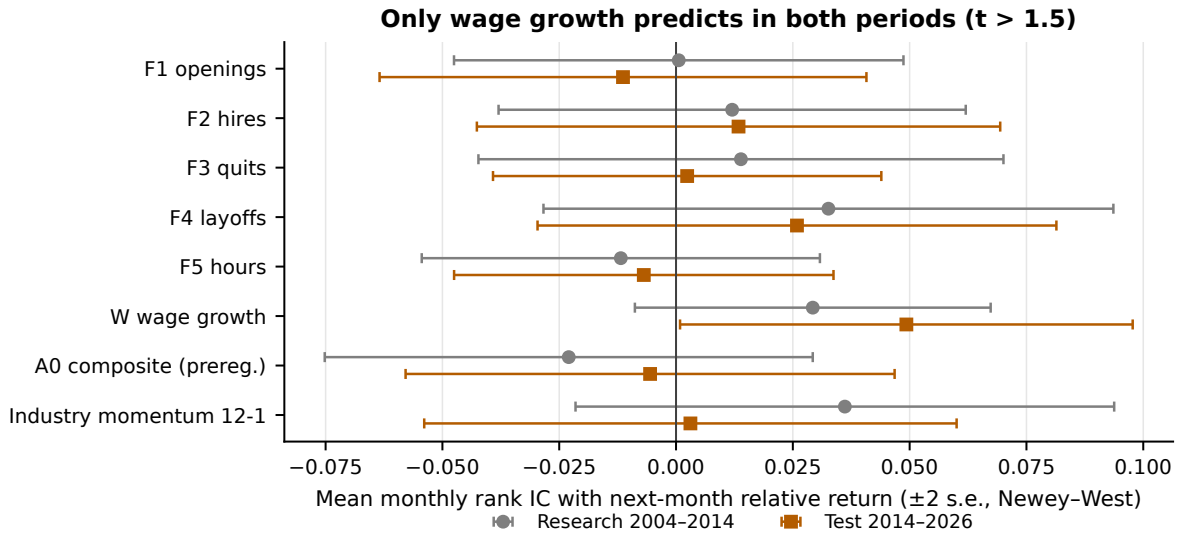


Figure 6: Wage growth is the only signal with a positive IC above 1.5 standard errors in both periods; layoffs and hours carry the opposite sign to the one the fixed rule imposed. *post hoc*

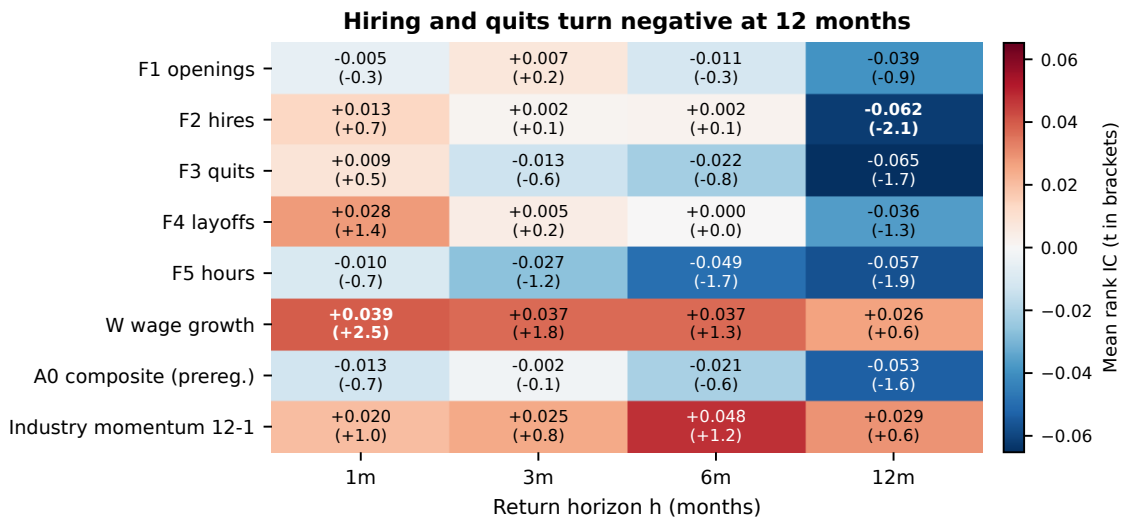


Figure 7: Hires and quits turn negative at 12 months ($t = -2.1, -1.7$): the investment channel shows up slowly, not within a month. *post hoc*

The agent experiment: design and results

The design is shown in Figure 2 of the main text; the research IC against the test IC of every feature is in Figure 4.

Table 15: Agent experiment: design parameters and counts in both studies.

confirmatory

study	item	value
Both	Arms	communicating (treatment), independent (control)
Both	Agents per arm	Temporal, Composition, Interactions
Both	Proposals per agent x repetitions	6 x 3
Both	Migration reports	after proposals 2 and 4, at most 120 words
Both	Selection (repetition 1 only)	$t \geq 1.5$, same sign in both halves, $ \text{corr} < 0.8$, at most 3 per arm
Study 1	Valid proposals	98 / 108
Study 1	Migration reports admitted	7 / 18
Study 1	Defined LLM-judge uptake labels	8 / 36
Study 1	Human-audited judge pairs	8
Study 1b	Valid proposals	106 / 108
Study 1b	Migration reports admitted	15 / 18
Study 1b	Defined LLM-judge uptake labels	84 / 88
Study 1b	Human-audited judge pairs	17
Study 1b	Human-judge agreement	10 / 17 (human yes 3, judge yes 10)

Table 16: Agent features: how many passed the research filter and the test criterion.

post hoc

	Repetition-1 candidates	Study 1	Study 1b
	valid candidates	33.0	35.0
	regime-gated	9.0	6.0
	test IC undefined (≤ 3 active months)	4.0	2.0
	corr(research IC, test IC)	0.22	0.37
	passed research filter	4.0	7.0
	selected	4.0	4.0
	selected and regime-gated	2.0	2.0
	selected with positive test IC	0.0	2.0

Table 17: Selected features, decoded after the test: research and test IC, months active.

post hoc

Study	Arm	Feature (decoded)	Research IC (n)	Test IC (n)
Study 1	independent	regime(yoy(Earnings), "low")	+0.065 (48)	n/a (3)
Study 1	independent	regime(spread(tsz(Layoffs, 60), tsz(Hires, 60)), "low")	+0.070 (43)	n/a (3)
Study 1	communicating	spread(chg(Layoffs, 3), chg(Openings, 3))	+0.037 (123)	-0.023 (142)
Study 1	communicating	spread(chg(Layoffs, 6), chg(Openings, 6))	+0.052 (120)	-0.004 (142)
Study 1b	independent	mult(regime(tsz(Hires, "exp"), "high"), csrank(tsz(Openings, "exp")))	+0.088 (15)	-0.028 (133)
Study 1b	communicating	tsz(lchg(lag(Earnings, 1), 12), 36)	+0.064 (78)	+0.022 (142)
Study 1b	communicating	regime(spread(chg(Quits, 3), chg(Hires, 3)), "low")	+0.125 (48)	n/a (3)
Study 1b	communicating	tsz(lchg(lag(Hours, 1), 12), 36)	+0.057 (78)	+0.011 (142)

The agents' formula grammar

Every proposal is a formula built from the 14 operations below (GRAMMAR in Study 1's `params.py`). The inputs are the six anonymised variables V01 to V06 (openings, hires, quits and layoffs rates, hours and earnings, in an order the agents never saw) and tightness T .

Table 18: The 14 operations of the agents’ grammar.*confirmatory*

Operation	What it computes
1 lag(x, k)	value of x , k months earlier ($k = 1, 2, 3, 6, 12$)
2 mavg(x, w)	moving average of x over w months ($w = 3, 6, 12$)
3 chg(x, k)	change: $x - \text{lag}(x, k)$
4 lchg(x, k)	log change: $\log x - \log \text{lag}(x, k)$
5 yoy(x)	year-over-year change: $x - \text{lag}(x, 12)$
6 accel(x)	acceleration: $\text{chg}(\text{chg}(x, 3), 3)$
7 persist(x, k)	share of months with a positive one-month change over the last k months ($k = 3, 6$)
8 tsz(x, w)	time-series z-score: (x minus its rolling or expanding mean) over its standard deviation ($w = 36, 60$ or expanding with a 36-month minimum)
9 csrank(x)	cross-sectional rank: average fractional rank across the 13 groups
10 ratio(x, y)	x/y (nonzero denominator required)
11 spread(x, y)	difference of cross-sectional z-scores of x and y
12 mult(x, y)	product $x \cdot y$
13 withT(x)	x interacted with tightness T
14 regime(x, s)	x kept only in one tightness regime ($s = \text{"low"}$ or "high")

Every proposal must also respect five constraints: a nesting depth of at most three; at most two variables plus T ; logarithms only of strictly positive values; rolling windows only with a complete history; and no arithmetic or operation beyond the fourteen above.

Study 1b: exploratory follow-up with Opus

After the test had been opened, the agent experiment was rerun with Claude Opus 5.5 at extra-high effort, with fresh agents and the same evaluator, selection rule and portfolio. Because the test period was already known to the team, the run is exploratory. Its research-selected primary, A4 (agent features, communicating), earned a test Sharpe of +0.01 (research −0.40) with an IC of −0.002 ($t = -0.10$), an alpha t of 0.05, a placebo p of 0.56 and a Sharpe of −0.19 at 25 bp; it failed gates G2 to G5 and had a Sharpe of −1.41 over the last 18 months. Its gross return came from one long position: durable manufacturing contributed +0.27 of cumulative gross return (sum of monthly contributions, in units of the book) in a group that is 60% semiconductors. Its A3 earned −0.46. The follow-up is evidence about agents, not about the strategy. A stronger model followed the protocol better and found no more signal: 106 of 108 valid proposals against 98, 15 of 18 admitted migration reports against 7, 84 of 88 usable judge labels against 8 of 36, yet no selected feature survived the test criterion in either study.

Appendix C. Study 2 in full: the post-hoc iteration

The main text reports Study 2 on data as known (Study 2.3). Studies 2.1 and 2.2 were first run on revised data with fixed publication lags; both versions are reported below.

Specifications, hashes and look counts

Table 19: Every Study 2 stage was written, hashed and committed before it ran. Times are UTC.*post hoc / forward test*

Specification (analysis/specs/)	SHA-256	Frozen	First run	Looks counted in the deflated Sharpe
study2_1_variants.md	ca658234	3 Oct, 09:00:13	09:01:55	112: about 108 diagnostic looks already taken, plus the 4 variants
forward_spec.md	735a8ffa	3 Oct, 09:03:56	book 09:17:55	none (forward test)
forward_amendment_001.md	c76cde86	3 Oct, 09:16:11	book 09:17:55	none
study2_2_hq12_robustness.md	312aef1d	3 Oct, 09:52:27	09:53:32	132: 112 plus 20 battery backtests
study2_3_robustness.md	068a3e03	4 Oct, 09:18:39	09:26:27	162: 132 plus 30 (24 as-known, 4 cap, 2 rank-weighted)

The run logs (analysis/output/study2/v2_run_log.md, v3_run_log.md, study2_3_run_log.md) hold the SHA-256 of every result table at its first run; the notebook recomputes them on each run and stops if one differs. Result files of Studies 2.1 and 2.2 keep the codes V2a to V2d inside (Appendix A).

Study 2.1: the four variants

Table 20: Post-hoc variants on revised data: net Sharpe by window and test statistics at 10 bp. Test figures are the test slice of a continuous run from May 2004 (A3's ledger starts from an empty book in November 2014). IC, alpha t and placebo are one-month statistics with 6 Newey–West lags; DSR counts 112 looks as independent trials, a rough deflation since the looks are correlated. *post hoc*

Variant	Research	Test	Full	Post-2010	Last 18m	Test IC (t)	Alpha t	Placebo p	DSR
W	+0.09	+0.16	+0.15	+0.12	−1.71	+0.049 (+2.04)	+0.29	0.02	0.02
SC	−0.06	−0.15	−0.11	−0.16	−1.34	+0.042 (+1.65)	−0.84	0.05	0.00
W+MOM	+0.27	+0.35	+0.32	+0.24	−0.78	+0.032 (+1.21)	+1.36	0.26	0.09
HQ12	+0.50	+0.29	+0.35	+0.30	+0.29	−0.020 (−0.87)	+1.37	0.85	0.06

The reading rule declared with the variants: positive Sharpe at 25 bp, IC $t \geq 2$, alpha $t \geq 2$, positive IC in both test halves, placebo $p < 0.05$ and DSR above 0.95; none of the four meets it. HQ12's pre-declared one-month placebo p is 0.85; a horizon-matched placebo added after the run gives 0.06 in the test and 0.04 over the full sample.

Study 2.2: the HQ12 battery on revised data

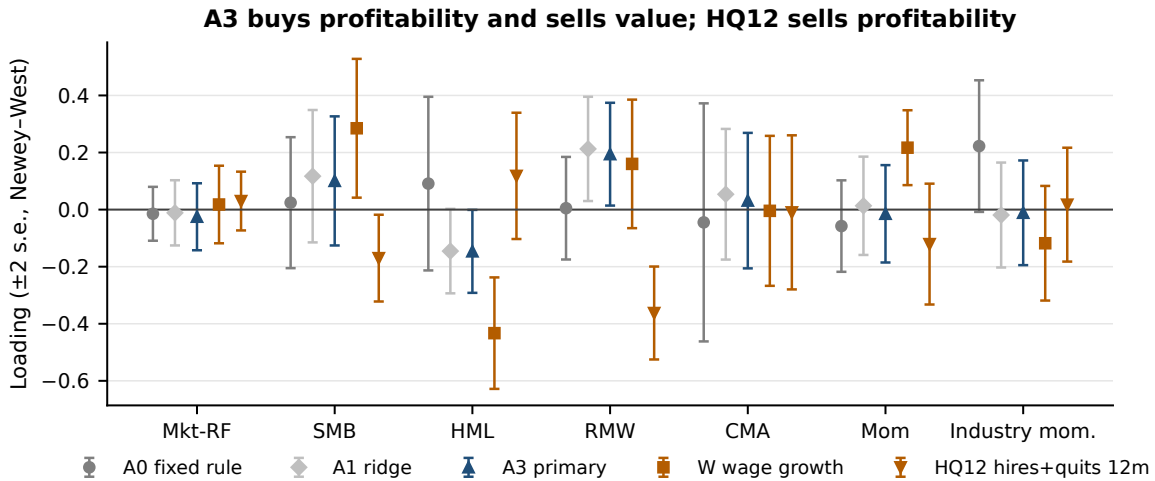


Figure 8: Factor loadings of the long–short books (Section 3.1): A3 buys profitability and sells value; HQ12 sells profitability. confirmatory (A0, A1, A3) / *post hoc* (W, HQ12)

Table 21: R3 holding period: Sharpe by window. *post hoc*

Holding	research	test	full	post-2010	last 18m	Test turnover
6 months	+0.31	+0.05	+0.15	+0.05	+0.50	0.49
9 months	+0.43	+0.14	+0.24	+0.16	+0.29	0.40
12 months	+0.50	+0.29	+0.35	+0.30	+0.29	0.37
18 months	+0.58	+0.30	+0.38	+0.34	+0.31	0.28

Table 22: R4 sub-periods.*post hoc*

Window	Months	Sharpe	Net p.a.	IC 12m (t)
test: first half	71	−0.21	−1.4%	+0.097 (+1.75)
test: second half	71	+0.64	+6.5%	+0.095 (+1.60)
test: excl. Mar 2020-Dec 2021	120	+0.37	+3.0%	+0.105 (+2.23)
test: pre-2020	62	+0.14	+0.8%	+0.114 (+1.92)
test: 2020 onward	80	+0.37	+3.9%	+0.080 (+1.54)
full: first half	134	+0.33	+2.0%	+0.038 (+0.73)
full: second half	134	+0.38	+3.3%	+0.107 (+2.66)
full: excl. Mar 2020-Dec 2021	246	+0.40	+2.9%	+0.073 (+1.93)
full: pre-2020	188	+0.36	+2.2%	+0.068 (+1.57)

Table 23: R5 costs: Sharpe by cost level and break-even.*post hoc*

Window	0 bp	10 bp	25 bp	50 bp	Break-even (bp)
research	+0.58	+0.50	+0.39	+0.20	76
test	+0.34	+0.29	+0.22	+0.09	68
full	+0.41	+0.35	+0.26	+0.11	69
post-2010	+0.35	+0.30	+0.21	+0.07	63
last 18m	+0.32	+0.29	+0.24	+0.16	99

Table 24: R6 factor regression of HQ12 with 12 Newey–West lags, revised data: loadings (*t* in brackets).*post hoc*

	Test	Full sample
Alpha (per month)	+0.26% (+1.48)	+0.26% (+1.92)
Market	+0.03 (+0.58)	−0.00 (−0.08)
SMB	−0.17 (−2.24)	−0.02 (−0.31)
HML	+0.12 (+1.07)	+0.10 (+1.23)
RMW	−0.36 (−4.45)	−0.16 (−1.83)
CMA	−0.01 (−0.07)	−0.09 (−0.87)
Momentum	−0.12 (−1.14)	−0.03 (−0.51)
Industry momentum	+0.02 (+0.17)	+0.06 (+0.93)

Table 25: R7 drop one group, every run.*post hoc*

industry	dropped group	test sharpe	full sharpe
Mining & logging	1	+0.33	+0.25
Construction	2	+0.30	+0.34
Durable mfg	3	+0.30	+0.30
Nondurable mfg	4	+0.40	+0.36
Wholesale	5	+0.36	+0.37
Retail	6	+0.29	+0.35
Transport & utilities	7	+0.30	+0.34
Information	8	+0.41	+0.54
Finance & insurance	9	+0.23	+0.23
Real estate	10	+0.48	+0.43
Prof. & business svcs	11	+0.26	+0.33
Health care	12	+0.18	+0.36
Accommodation & food	13	+0.32	+0.28

Table 26: R9 block-bootstrap 90% interval.*post hoc*

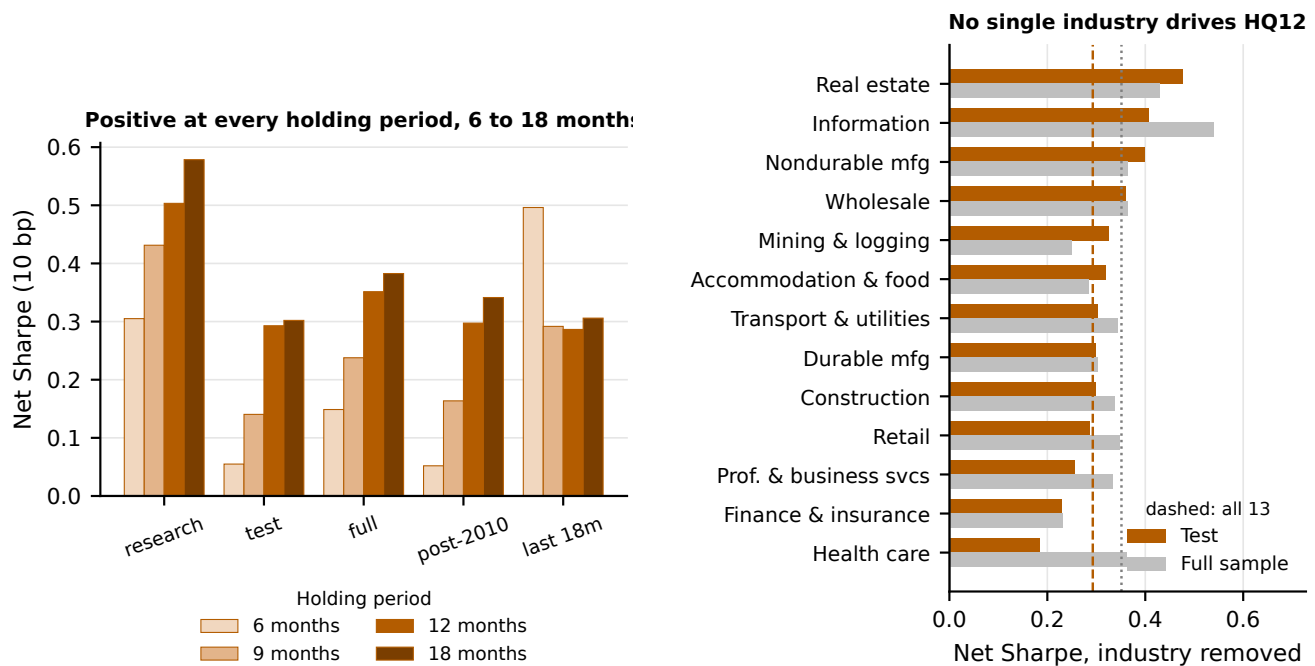
Window	Sharpe	90% interval	Months	Excludes 0
test	+0.29	[-0.07, +0.65]	142	no
full	+0.35	[+0.05, +0.64]	268	yes

Table 27: R11 fundamental law in the test window: IC, breadth, transfer coefficient, implied and realised ratio, revised data. confirmatory (A3) / *post hoc*

Book (test window)	IC	Horizon	BR/yr	TC	Implied IR	Ceiling (TC=1)	Realised Sharpe
A3 primary	-0.021	1m	156	0.80	-0.21	-0.27	-0.58
W wage growth	+0.049	1m	156	0.91	+0.56	+0.62	+0.16
HQ12 -(hires+quits) 12m	+0.096	12m	13	0.48	+0.17	+0.35	+0.29

Table 28: The pre-declared reading of the battery on revised data: HQ12 is fragile on R4.*post hoc*

condition	holds
R1 hires-only and quits-only test Sharpe > 0	yes
R2 test 12m IC $t \geq 2$ and 1m IC $t < 2$	yes
R3 holding 9 and 18 months test Sharpe > 0	yes
R4 both test halves Sharpe > 0	no
R5 test break-even cost > 25 bp	yes
R7 minimum drop-one test Sharpe > 0	yes
R9 90% interval excludes 0 (test or full)	yes

**Figure 9:** Left: HQ12 is positive at every holding period from 6 to 18 months. Right: no single industry carries HQ12; removing any one leaves its test Sharpe between +0.18 and +0.48 on revised data (+0.05 and +0.36 as known). *post hoc*

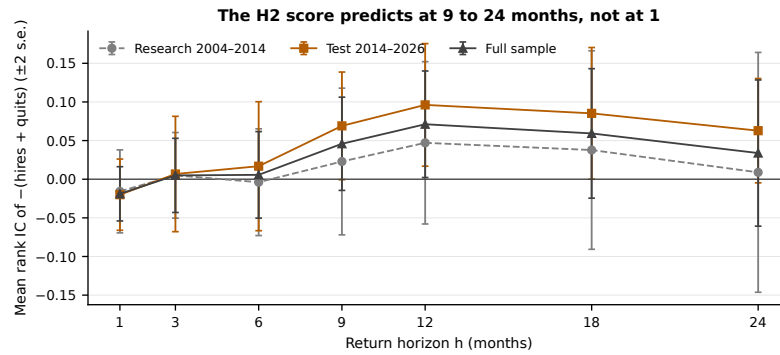


Figure 10: Horizon profile of the HQ12 score in every window: no one-month IC, positive from 9 to 24 months. *post hoc*

Study 2.3: as-known rerun, risk overlay, rank weights, agents, capacity

The as-known inputs come from Alex's data package. Before any variant was rerun we rebuilt his raw features from the as-known panel and his A0 score from his standardised features: both match his saved files to machine precision.

Table 29: R1: the four variants on fixed-lag and on as-known data, test window. The last two columns are the share of months in which both versions rank the same three groups highest and lowest. *post hoc*

Variant (test window)	Sharpe, fixed lag	Sharpe, as known	IC (t), fixed lag	IC (t), as known	Alpha t	Same top 3	Same bottom 3
W	+0.16	+0.13	+0.049 (+2.04)	+0.041 (+1.73)	+0.18	76%	70%
SC	-0.15	+0.24	+0.042 (+1.65)	+0.021 (+0.75)	+0.71	0%	0%
W+MOM	+0.35	+0.46	+0.032 (+1.21)	+0.036 (+1.40)	+1.93	84%	74%
HQ12	+0.29	+0.20	+0.096 (+2.43)	+0.085 (+2.26)	+0.92	39%	51%

Table 30: R1: HQ12 battery, fixed lag against as known, every row; the source of Table 3.*post hoc*

HQ12 check	Fixed lag (Study 2.2)	As known (Study 2.3)
Test Sharpe (10 bp)	+0.29	+0.20
Full-sample Sharpe (10 bp)	+0.35	+0.31
12-month IC, test	+0.10	+0.08
12-month IC t, test	+2.43	+2.26
1-month IC t, test	−0.87	−0.42
Component hires only, test Sharpe	+0.19	+0.19
Component quits only, test Sharpe	+0.16	+0.17
Holding 9 months, test Sharpe	+0.14	+0.16
Holding 18 months, test Sharpe	+0.30	+0.11
First test half, Sharpe	−0.21	−0.23
Second test half, Sharpe	+0.64	+0.49
Test Sharpe at 0 bp	+0.34	+0.25
Test Sharpe at 25 bp	+0.22	+0.13
Test Sharpe at 50 bp	+0.09	−0.00
Break-even cost, test (bp)	68	50
Drop one group, minimum test Sharpe	+0.18	+0.05
Drop one group, maximum test Sharpe	+0.48	+0.36
Bootstrap 90% interval, test: low	−0.07	−0.14
Bootstrap 90% interval, test: high	+0.65	+0.57
Alpha per month (%), test, HAC 12	+0.26	+0.16
Alpha t, test, HAC 12	+1.48	+0.92
Bootstrap 90% interval, full: low	+0.05	+0.02
Bootstrap 90% interval, full: high	+0.64	+0.60
Alpha per month (%), full, HAC 12	+0.26	+0.21
Alpha t, full, HAC 12	+1.92	+1.81
Deflated Sharpe, test (132 looks fixed lag; 162 as known)	+0.05	+0.02

Table 31: R2: ex-ante volatility of the unit book and gross exposure needed to reach 10%: 5th, 50th and 95th percentile over the test months.*post hoc*

book	unit vol p5	unit vol p50	unit vol p95	gross needed p5	gross needed p50	gross needed p95
A3	0.045	0.075	0.110	1.85	2.60	4.19
HQ12	0.033	0.064	0.103	1.79	2.45	4.24

Table 32: R3: top 3 / bottom 3 against rank-weighted books, test window. TC is the transfer coefficient; the implied IR is $IC \times \sqrt{BR} \times TC$.*post hoc*

Book	Weights	Test Sharpe	TC	Turnover	Implied IR
A3	top 3 / bottom 3	−0.58	0.80	0.91	−0.21
A3	rank-weighted	−0.49	0.83	0.86	−0.22
HQ12	top 3 / bottom 3	+0.29	0.48	0.37	+0.17
HQ12	rank-weighted	+0.31	0.49	0.37	+0.17

Table 33: Turnover and costs in the test window, revised data. Break-even is the industry cost at which the net return is zero. *confirmatory (A3) / post hoc*

Book (test window)	Turnover p.a.	Cost drag p.a.	Gross p.a.	Net p.a.	Break-even (bp)
A3 agent features, independent (primary)	11.0x	1.11%	-3.5%	-4.6%	-32
W wage growth (declared headline)	4.7x	0.47%	+2.1%	+1.7%	+46
SC signed composite	8.7x	0.88%	-0.5%	-1.3%	-5
W+MOM wage growth + momentum	5.8x	0.59%	+3.6%	+3.1%	+62
HQ12 hires + quits reversal, 12-month hold	4.4x	0.44%	+3.0%	+2.6%	+68
A3 rank-weighted	10.3x	1.05%	-2.5%	-3.6%	-24
HQ12 rank-weighted	4.4x	0.45%	+3.0%	+2.6%	+69

Table 34: R4: agent proposals by study, arm and seed: valid proposals, best and mean research IC, eligible features (seed 1 only) and share with a regime gate. *post hoc*

Study	Arm	Seed	Valid	Best IC	Mean IC	Eligible	Regime-gated
Study 1	independent	1	17/18	+0.101	+0.014	2	29%
Study 1	independent	2	17/18	+0.102	-0.006		24%
Study 1	independent	3	13/18	+0.046	-0.014		15%
Study 1	communicating	1	16/18	+0.052	-0.003	2	25%
Study 1	communicating	2	17/18	+0.058	+0.005		6%
Study 1	communicating	3	18/18	+0.037	-0.008		22%
Study 1b	independent	1	18/18	+0.088	-0.002	1	28%
Study 1b	independent	2	18/18	+0.045	-0.006		11%
Study 1b	independent	3	18/18	+0.045	+0.002		6%
Study 1b	communicating	1	17/18	+0.125	+0.026	6	6%
Study 1b	communicating	2	18/18	+0.191	+0.049		39%
Study 1b	communicating	3	17/18	+0.091	+0.009		47%

Table 35: R4: arm gap against seed spread, exact permutation test over 20 relabelings. *post hoc*

Study	Metric	Communicating minus independent	Largest seed range	Gap within seed range	Permutation p
Study 1	best IC	-0.034	0.056	yes	0.30
Study 1	mean IC	-0.001	0.028	yes	1.00
Study 1b	best IC	+0.076	0.099	yes	0.10
Study 1b	mean IC	+0.030	0.040	yes	0.10

Table 36: R5: tokens per proposal call by arm. *post hoc*

Study	Arm	Calls	Tokens, total	Tokens per call	Output tokens per call
Study 1	independent	54	120,932	2,239	183
Study 1	communicating	54	134,185	2,485	193
Study 1b	independent	54	180,555	3,344	1,257
Study 1b	communicating	54	214,003	3,963	1,509

Table 37: R5: Spearman correlation between a proposal’s output tokens and its research IC. *post hoc*

Study	Proposals	Spearman, output tokens vs IC	Spearman, output tokens vs absolute IC
Study 1	98	+0.13	+0.21
Study 1b	106	+0.41	+0.24

Table 38: R6: participation ratio of the 13-group correlation matrices.*post hoc*

matrix	participation ratio
13 group excess returns	2.0
13 group relative returns (group minus equal-weight mean)	9.1
z(W) across groups	7.4
HQ12 score across groups	10.1

Table 39: R7: capacity with the proxy ETFs: the binding ETF, its average daily dollar volume, and the assets at which monthly trading reaches 1% and 5% of it.*post hoc*

Book	Binding ETF	ADV, USD m	Trade / assets	Assets at 1% ADV	at 5%	SPY ADV, USD bn
A3	PEJ	3.2	6.3%	0.5	2.5	47
HQ12	PEJ	3.2	2.7%	1.2	5.9	47

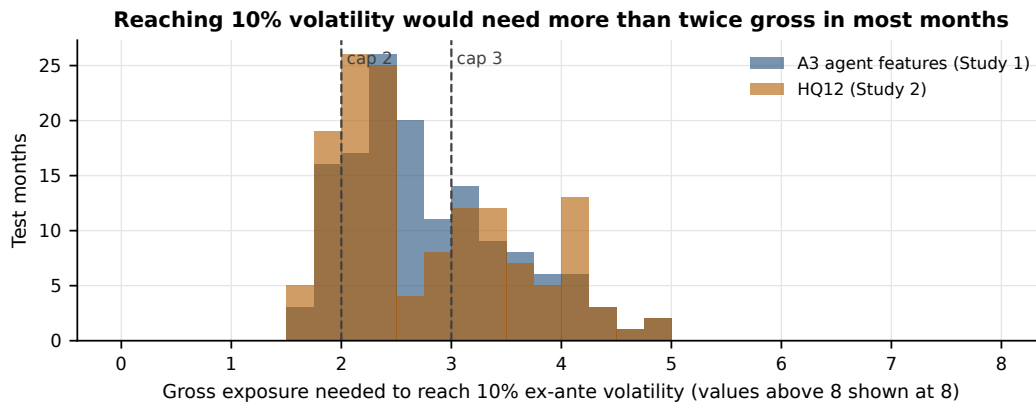
Table 40: R8: judge with a verbatim quote on the 25 hand-labelled pairs. “Usable” counts JSON-only answers; the third row of each study accepts JSON wrapped in a code fence and is outside the pre-declared rule.*post hoc*

Study	Judge	Pairs	Usable	Agree with human	Judge yes	Human yes	False yes	Missed yes
Study 1	original judge	8	2	1	1	1	1	0
Study 1	judge with verbatim quote	8	8	7	2	1	1	0
Study 1	with quote, fenced JSON accepted (supplementary)	8	8	7	2	1	1	0
Study 1b	original judge	17	17	10	10	3	7	0
Study 1b	judge with verbatim quote	17	7	3	5	3	4	0
Study 1b	with quote, fenced JSON accepted (supplementary)	17	17	9	11	3	8	0

Table 41: R9: feedback ablation, independent arm, research data only, with and without the tightness-tercile split in the feedback. “Informative” counts months in which a proposal’s cross-section is not constant.*post hoc*

Feedback	Seeds	Proposals	Valid	With a regime gate	Share	Informative in under half of months	Best research IC
with tercile split	3	54	48	4	8%	8%	+0.070
without tercile split	3	54	50	3	6%	6%	+0.070

The ablation rebuilt the agents’ research inputs from Alex’s package; they reproduce 94 of the 98 saved feedbacks exactly, and the largest difference is 0.11 in a t -statistic. The judge calls for Study 1b pairs used a newer command-line client than the Study 1 pairs because the older one does not serve that model; calls that failed in transport were repeated once, and every call is in the log (analysis/output/study2/11m/).

**Figure 11:** Reaching 10% volatility would need more than twice gross in most test months, so the cap of 2 binds. *post hoc*

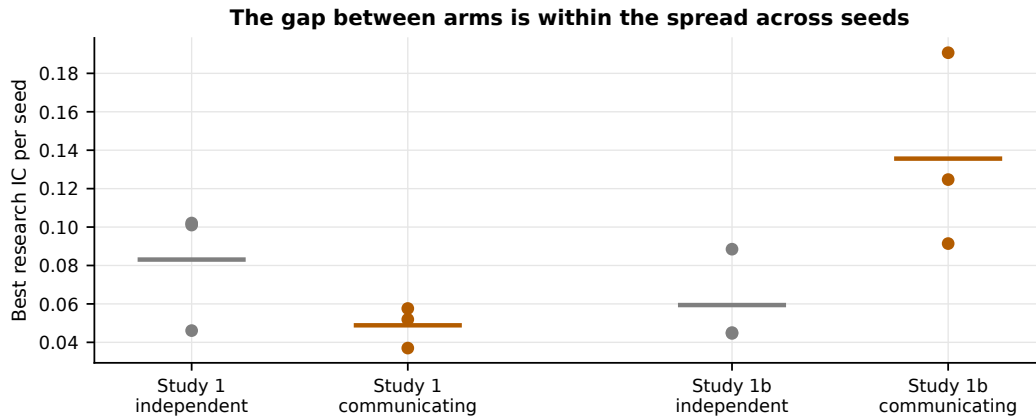


Figure 12: Best research IC per seed: the gap between arms is within the spread across seeds in both studies (Section 3.4). *post hoc*

Appendix D. Forward test: frozen specification and Q4 2026 books

- analysis/specs/study2_1_variants.md: four post-hoc variants and a reading rule; SHA-256 ca658234. . . , frozen 2026-10-03T09:00:13Z, first run 09:01:55Z.
- analysis/specs/forward_spec.md: decision 30 September 2026, information cutoff 29 September, one tranche held to 31 December, 60-month beta hedge in SPY, 10 bp and 2 bp costs; SHA-256 735a8ffa. . . , frozen 09:03:56Z.
- analysis/specs/forward_amendment_001.md: the fixed-lag information set (JOLTS July 2026, CES August 2026, as published on 29 September) after the pre-committed check stopped the first run; W+MOM and HQ12 added after the Study 2.1 results were seen and before any Q4 return beyond 1–2 October; SHA-256 c76cde86. . . , frozen 09:16:11Z; book built 09:17:55Z.
- analysis/specs/study2_2_hq12_robustness.md: the HQ12 battery; SHA-256 312aef1d. . . , frozen 09:52:27Z, first run 09:53:32Z. analysis/specs/study2_3_robustness.md: SHA-256 068a3e03. . . , frozen 4 October 09:18:39Z, before its first output.
- Books: analysis/output/forward/book_2026Q4.csv (group weights for W, SC, W+MOM and HQ12), book_2026Q4_etf.csv (netted ETF weights), book_2026Q4_hedge.csv (SPY hedge per unit: -0.40 , -0.06 , -0.24 , -0.02), etf_proxies.csv. Holdings: W is long construction, durable manufacturing and information and short mining, health care and accommodation; SC is long durable manufacturing, retail and transport and short real estate, health care and accommodation; W+MOM is long construction, durable manufacturing and finance and short retail, transport and professional services; HQ12 is long finance, real estate and accommodation and short mining, durable manufacturing and retail. W+MOM's tradable book has four legs, because its long durable-manufacturing and short professional-services positions both map to the industrials ETF and cancel.
- Power: HQ12's test volatility is about 9% a year, so one quarter's return has a standard deviation near 5%; a Q4 return between -10% and $+10\%$ is within two standard deviations of zero. The book was first built at 09:17:55Z on 3 October: commit 3e8ecd7 (09:18:29Z) holds a byte-identical book_2026Q4.csv (SHA-256 857275. . .); book_first_build.sha256 records that first build and later notebook runs must reproduce it.
- Horizon: HQ12 predicts returns nine to 24 months ahead and has no IC at three months, so its Q4 return will mostly measure noise. The informative evaluation is the twelve-month return of the same tranche, to 30 September 2027 (Section 3.3).
- Evaluation: the ETF books from the 30 September close to the 31 December close after costs, and the same books on Ken French group returns once published; reported whatever they show. Cadence: a weekly tracker

was considered and deferred by team decision, since with one quarter of data and no capital at risk a weekly readout adds noise without adding power (Section 3.3). We will evaluate once at the 31 December close.

Appendix E. Claude, interaction by interaction

Condensed from the team’s written evaluation. Paths are relative to `study1_preregistered/outputs/follow_the_workers/` unless stated; Study 1b has the same files under `follow_the_workers_opus55_xhigh/`.

Claude took every AI role: Sonnet 5.5 for Study 1, Opus 5.5 at extra-high effort for Study 1b, and later Claude models for the post-mortem and the iteration. This term’s course allows it; the switch from the ChatGPT template is logged as preregistration amendments 001 and 002 (`outputs/prereg_amendments/`, `outputs/deviations.md`), and the log file keeps the template name `chatgpt_log.md` with Claude content.

Table 42: Claude, interaction by interaction: what it got right, what it got wrong, what we did. The detail of each interaction and the transcripts follow the table.

Interaction	What it got right	What it got wrong	What we did
P1 audit (Sonnet)	Flagged every real gap in the mapping	5 of 25 findings unsupported; design choices called “confirmed errors”	Tested all 25; one metadata fix
P2 agents (Sonnet; Opus in 1b)	Blind protocol followed; 98 and 106 of 108 proposals valid	No feature survived the test; A3’s two features active in 3 months	Mechanical selection; no proposal edited
P3 judge	Never missed a true reuse	7 false yeses in 17 audited pairs	Hand audit; quote requirement tested in Study 2.3
P4 red team (Sonnet)	6 of 10 concerns matched registered tests	A p -value never computed; new thresholds after seeing results	Registered tests kept; new thresholds rejected
P5 post-mortem [†]	Traced the failure to features active in 3 months	Fixes read off the test results	Became Study 2, labelled post hoc
P6 iteration [†]	Reproduced Study 1 exactly; froze every stage	Wrong assumption on a JOLTS release date	Its own check stopped the run; amendment disclosed

P1, data and code audit The system prompt made Claude “a careful data and code auditor” with no tools; the prompt in Section 2.3 was followed by the manifest, the crosswalk and the as-of code (38,717 characters). It returned 25 numbered findings, all tested (`outputs/p1_review.md`): five were claims the files did not support, among them that group 10’s CES series “silently vanishes” (584 shared, nonmissing rows were present); one led to a metadata fix (the crosswalk now separates the prescribed match from the verified French scope); the rest were documented design choices or limitations, and none was a look-ahead error.

P2, signal agents Six agents (three independent, three communicating) over three seeds proposed six features each in a 14-operation grammar (Table 18), blind to names and dates, with research-period feedback only. 98 of 108 proposals passed validation; the blinding audit of all 199 calls passed (108 proposals, 18 migration reports, one syntax repair). The communicating agents exchanged migration reports capped at 120 words; 11 of 18 were rejected, 10 of them for running to 121–133 words although the prompt asked the agent to check its count. Selection was mechanical, two per arm from seed 1 only, and no engineer edited a proposal. Both independent-arm picks, decoded after the test, were low-tightness gates: `regime(yoy(Earnings), "low")` and `regime(spread(tsz(Layoffs, 60), tsz(Hires, 60)), "low")`, active in 3 of 142 test months. Our first critique blamed the regime split in the agents’ feedback; removing it in Study 2.3 did not remove the gates (Table 41, Section 3.4).

P3, judge In Study 1b, items 2, 3 and 13 cite “submission 2’s significant negative IC”, a result that exists in another report; item 12 cites “submission 4” from a report with two submissions. Items 8, 16 and 17 are judgment calls (a methodological tip reused; a report’s topic shared but its result contradicted). The same proposal paired with two reports (items 12 and 17) got a yes both times, which is what a judge that matches on vocabulary would do. With 17 dependent pairs the sample is small. In Study 1 only 8 of 36 judge outputs were usable and 8 pairs were audited. The judge said “uses a claim” 10 times where the human said 3, so its 62% uptake rate overstates communication between agents. Because every error was a false yes, our critique proposed requiring the judge to quote the exact claim it relies on; in Study 2.3, the quote fixed Sonnet’s format but not Opus’s false yeses (Section 3.4).

P4, red team Claude received the research-period result tables, without dates or names, and was asked for ten reasons the results could be spurious, each with a test and a “do not implement” criterion (prompt in Section 2.3). It read the tables correctly: six of its ten concerns (the placebo, 25 bp costs, the two test halves, tightness and the data versions) were already covered by registered diagnostics, which confirms that the preregistration had anticipated the main risks. For the other four, no additional test was adopted. Its errors were of two kinds. It asserted a permutation p -value “already” above 0.10 that nobody had computed, and it proposed new pass/fail thresholds (a 95th-percentile gate, a deflated Sharpe ratio of 0.95 and a rolling 24-month rule) after seeing the research tables. We kept the registered tests and rejected the new thresholds: adopting them would have let results already seen shape the rules used to judge them. A red team is useful as a checklist against the registered tests, not as a source of new criteria once results are visible.

P5, post-mortem review After the test had been opened, Claude reviewed Study 1’s results and diagnostics. Its diagnosis was right: it traced the primary strategy’s failure to two agent features active in only 3 of 142 test months, and it identified the wrong signs, the wrong horizon and the untestable tightness regime. Its three proposals (bring back wage growth, let the research data set the signs, look at a twelve-month horizon) became Study 2. But every one of them was read off the test results, so they are post hoc by construction, and the signal it called stable, wage growth, then lost heavily over the last 18 months (−1.31 on data as known). The review also contained errors of its own (a data-vendor guess from a code path, a momentum definition that included the latest month, wage ICs computed from a non-frozen start date), recorded with their corrections in docs/FINDINGS_AND_PROPOSAL.md. We used the diagnosis, labelled every proposal post hoc, froze each stage of Study 2 before running it, and left the final judgment to the forward test.

P6, iteration agent Given task prompts for each stage and a one-page rules file, the iteration agent rebuilt the analysis, checked Study 1’s folder against its fingerprint, reproduced Study 1’s sealed returns to 10^{-16} before running anything new, and ran Study 2 in three frozen stages with a run log that stores the hash of every result table. Every number in the report carries a source comment. When it assumed that August JOLTS data would not be published by 29 September, the pre-committed data-release check stopped the forward-test build; the fix is disclosed as forward amendment 001.

Bottom line Every Claude output needed a human check: 25 P1 dispositions, 10 P4 dispositions and 25 hand-labelled migration pairs (8 in Study 1, 17 in Study 1b).

E.1 Transcripts and verbatim excerpts

- P1: prompt outputs/p1_request.json; response outputs/p1_response.md; dispositions outputs/p1_review.md: 25 findings, 5 not supported by the files, 1 metadata fix, 19 documented design choices, limitations or checks that found no error.

- P2: system and turn prompts in `params.py` (SYSTEM_PROMPT, TURN_PROMPT); every call with its raw response in `outputs/agents/log.jsonl`; proposals in `outputs/candidates.csv`; selected features decoded in `analysis/output/study1/tables/d7_selected_features.csv`.
- P3: migration reports in `outputs/agents/migrations.json`; the judge prompt in `agents.py`, function `trace_migrations`; judgments in `outputs/agents/migration_judgments.json`; the human audit in `outputs/human_review_submission.json` (Study 1b also has a `.md` version).
- P4: prompt outputs/`p4_request.json`; response outputs/`p4_response.md`; dispositions outputs/`p4_review.md`.
- P5: the review and its output, `docs/FINDINGS_AND_PROPOSAL.md`, with the numbers in `analysis/output/study1/diagnostics.md`.
- P6: task prompts for each stage of the iteration (rebuild and diagnostics; Study 2.1 and 2.2 freezes, forward test and report draft; repository passes; Study 2.3 and this rewrite) and a one-page rules file (Study 1 folder read-only, one label per result, specifications hashed before they run, a source comment on every number, stop at a failed gate), kept outside the repository; outputs in `analysis/run.ipynb`, `analysis/output/`, `results/` and this report.
- Study 2.3 calls (judge with a quote; feedback ablation): every prompt and raw response in `analysis/output/study2/llm/r8_calls.jsonl` and `r9_calls.jsonl`; code in `analysis/llm_runs.py`.

Verbatim excerpts Short excerpts, copied from the files above without editing.

P1 output, opening of the response

outputs/p1_response.md

Scope: only the supplied material. I did not verify live FRED data. I mark each item as confirmed (visible in the text) or as a question or limitation. Confirmed errors: mapping and manifest. 1. Manifest rows for CEU5500000007 and CEU5500000008 with group 10 duplicate the group 9 rows. The same series ID is mapped to two groups, and group 10 (Real estate) has no distinct CES series. [...] Test: group the manifest by id and assert one group per series. Compare the feature columns of groups 9 and 10, which will be identical.

P2 output, one agent proposal, decoded after the test

independent arm, seed 1, Temporal agent, proposal 3

Expression regime(yoy(V05),"low"). Hypothesis: groups with higher year-over-year change in V05 (dollars per hour) will have higher next-month returns in low-T regimes, because accelerating compensation signals stronger conditions not yet fully priced in. The effect should vanish in high T. [V05 was later decoded as average hourly earnings; the feature was selected for A3 and was active in 3 of 142 test months.]

P3 judge versus human, two audited pairs

outputs/human_review_submission.md, Study 1b

Item 5, human yes, judge yes: "The report found that V02 outpacing V03 led to significant underperformance, and that the V03-favoring reversal was positive across T regimes. The proposal uses that V03/V02 composition signal without conditioning on T." Item 2, human no, judge yes: "It says it reverses 'an observed negative signal,' but this report has no negative signal: the V02/V03 ratio had IC 0.01 and the V02-vs-V04 shift had IC 0."

P4 output, first of ten reasons

outputs/p4_response.md

1. The IC signals are pure noise (no predictive content). All t-statistics are below 1.3 and the signs flip across variants. Test: a block-bootstrap or permutation test of IC, shuffling signals within each date, 1000 or more draws, with Newey-West lags of 6. Do not implement if: the permutation p-value is above 0.10 for any variant. This is already the case, so the answer is do not implement. [Disposition in `p4_review.md`: covered by the registered circular-shift placebo; no new test adopted.]

Appendix F. Key code

As-of lookup The value of a series known on a cutoff date (Study 1 `data.py`).

```
def vintage_view(frame, cutoff, first=False):
```

```

cutoff = pd.Timestamp(cutoff).strftime('%Y-%m-%d')
if first:
    eligible = frame.drop_duplicates('observation_month', keep='first')
    eligible = eligible.loc[eligible.realtime_start <= cutoff]
else:
    eligible = frame.loc[(frame.realtime_start <= cutoff) & (frame.realtime_end >= cutoff)]
if eligible.duplicated('observation_month').any():
    raise ValueError('Overlapping vintage intervals')
return eligible.set_index('observation_month')[['value', 'realtime_start']].sort_index()

```

Signal construction Fixed-lag signals and tightness (analysis/data.py).

```

def signals(folder=P.FRED, last_decision=None, groups=None):
    panel = labor_panel(folder)
    avg3 = lambda x: x.rolling(3, min_periods=3).mean()
    raw = {}
    to_decision = lambda frame, lag: frame.set_axis(frame.index + lag) # observation o -> decision o+lag
    for i, m in enumerate(P.JOLTS_MEASURES, 1):
        x = avg3(panel[m])
        raw[f'F{i}'] = to_decision(x - x.shift(12), P.JOLTS_LAG)
    h, e = avg3(panel['H']), avg3(panel['E'])
    raw['F5'] = to_decision(np.log(h / h.shift(12)), P.CES_LAG)
    raw['W'] = to_decision(np.log(e / e.shift(12)), P.CES_LAG)
    # last decision month at which every feature has its lagged observation
    end = last_decision or min(v.dropna(how='all').index.max() for v in raw.values())
    decisions = pd.period_range(P.SIGNAL_START, end, freq='M')
    features = {}
    for name, value in raw.items():
        value = value[list(groups)] if groups is not None else value # drop-one-industry diagnostics
        z = standardize(value.reindex(decisions))
        z.index = z.index + 1 # earning month
        features[name] = z
    # T at decision d uses the latest month <= d-2 with both V and U observed (frozen rule); this
    # matters once: the Oct 2025 household survey (UNEMPLOY) was never collected.
    ratio = np.log(panel['V'] / panel['U']).ffill()
    t = to_decision(ratio, P.JOLTS_LAG).reindex(decisions)
    t.index = t.index + 1
    return features, t

```

Backtest core Tranche, hedge, scaling and costs (Study 1 data.py, excerpt).

```

for month, score in scores.iterrows():
    tranche = pd.Series(0., index=groups)
    if signal_available.loc[month]:
        rank = score.sort_index().sort_values(kind='stable')
        tranche.loc[rank.index[-P.PORTFOLIO['longs']:]] = 1/P.PORTFOLIO['longs']
        tranche.loc[rank.index[:P.PORTFOLIO['shorts']]] = -1/P.PORTFOLIO['shorts']
    tranches.append(tranche)
    book = sum(tranches[-holding:])/holding
    betas = pd.Series({g: np.cov(historical[g], market, ddof=1)[0,1]/variance for g in groups})
    raw = np.r_[book.to_numpy(), -float(book@betas)]
    covariance = LedoitWolf().fit(historical.to_numpy()).covariance_
    risk = np.sqrt(max(0, float(raw@covariance@raw))*12)
    if risk <= 0 and np.abs(raw).sum() > 1e-14:
        raise ValueError('Degenerate portfolio risk estimate')

```

```

scale = min(P.PORTFOLIO['vol_target']/risk,P.PORTFOLIO['gross_cap']/np.abs(raw).sum()) if risk>0 else 0.
weights = pd.Series(raw*scale,index=pretrade.index)
assert abs(weights[groups]@betas+weights['market'])<1e-10
assert weights.abs().sum()<=P.PORTFOLIO['gross_cap']+1e-10
turnover = float((weights[groups]-pretrade[groups]).abs().sum())
hedge_turnover = abs(weights['market']-pretrade['market'])
transaction = turnover*cost_bps/10000+hedge_turnover*P.PORTFOLIO['hedge_cost_bps']/10000
# Charge borrowed securities plus negative residual cash, at the stated annual rate.
borrow_base = -weights.clip(upper=0).sum()
finance_base = max(0,float(weights.sum()-1))
financing = (borrow_base+finance_base)*borrow_bps/10000/12
gross = float(weights[groups]@returns['excess'].loc[month,groups]+weights['market']*factors['Mkt-RF'])
net = gross-transaction-financing

```

Appendix G. Data sources and links

- Ken French Data Library: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html (49 industry portfolios, five factors, momentum; CRSP build 202608). Live, checked 3 October 2026.
- JOLTS by industry, not seasonally adjusted: FRED series JTU<code>JOR/HIR/QUR/LDR, for example <https://fred.stlouisfed.org/series/JTU2300HIR> (checked 3 October 2026, series runs through August 2026). Codes 110099, 2300, 3200, 3400, 4200, 4400, 480099, 5100, 5200, 5300, 540099, 6200, 7200.
- CES hours and earnings: CEU<code>07 and CEU<code>08 (construction earnings CES20000000008); codes 10000000, 20000000, 31000000, 32000000, 41420000, 42000000, 43000000, 50000000, 55000000, 60000000, 65000000, 70000000. Checked 3 October 2026 via FRED.
- Tightness: JTSJOL and UNEMPLOY. Vintages: <https://alfred.stlouisfed.org/> (live, checked 3 October 2026).
- As-known panels and saved outputs of Study 1 and Study 1b: Alex Roesler's data package (4 October 2026; 1,191 files checked against its inventory), read from analysis/data/alex_package/.
- ETF proxies and their caveats: analysis/output/forward/etf_proxies.csv. Daily dollar volume for the capacity note: Yahoo Finance, 252 trading days to 30 September 2026.

Appendix H. From 49 French industries to 13 labor groups

Table 43: The 13 groups: JOLTS code, Ken French members and the largest member's share of August 2026 market capitalisation. *confirmatory*

Group	JOLTS	Ken French industries	Largest member
Mining and logging	110099	Gold, Mines, Coal, Oil	Oil 79%
Construction	2300	Cnstr	100%
Durable manufacturing	3200	Toys, BldMt, Steel, FabPr, Mach, ElcEq, Autos, Aero, Ships, Guns, Hardw, Chips, LabEq, MedEq	Chips 60%
Nondurable manufacturing	3400	Food, Soda, Beer, Smoke, Hshld, Clths, Drugs, Chems, Rubbr, Txtls, Paper	Drugs 60%
Wholesale	4200	Whlsl	100%
Retail	4400	Rtail	100%
Transport and utilities	480099	Trans, Util	Util 61%
Information	5100	Telcm	100%
Finance and insurance	5200	Banks, Insur, Fin	Banks 51%
Real estate	5300	RIEst	100%
Professional and business services	540099	BusSv	100%
Health care	6200	HIth	100%
Accommodation and food	7200	Meals	100%

Groups 9 and 10 share one CES series, and groups 7, 12 and 13 use broader CES supersectors. The participation ratios of the 13-group correlation matrices put a number on the effective width: 2.0 for excess returns (the market dominates), 9.1 for relative returns, 7.4 for $z(W)$ and 10.1 for the HQ12 score.

Appendix I. Where each requirement is answered

Table 44: The brief and the rubric, line by line.

Requirement	Where	Page
1 Executive summary: one paragraph; strategy stated; “Do not implement” with justification	Section 1	1
2a Ideas: origin of the strategy	Section 2.1: motivation, related literature, Q1, Q2, H1–H3	1
2b Data: sources described	Section 2.2, Figure 1; links in Appendix G	2
2c Implementation: risk modeling	Section 2.3, “Portfolio and risk”, Equation (2); equations in Appendix B	4
2c Implementation: turnover and costs	Section 2.3, “Turnover and costs”; Appendix C	3
2c Implementation: at least three high-quality prompts	Section 2.3, “Prompts”: idea generation, coding support and robustness design	3
3a Style and risk factor exposures	Section 3.1; Figure 8	6
3b Time patterns: full sample, post-2010, recent 12–18 months	Section 3.2, Table 2	6
3c Robustness and limits	Section 3.3, Table 3, Figure 3; Appendix B, Appendix C	7
3d AI evaluation: usefulness, flaws, adaptations	Section 3.4, Table 4, Figure 4; Table 42	8
4 Appendices: AI transcripts	Appendix E, Appendix E.1	28
4 Appendices: tables and plots	Appendix B, Appendix C	13
4 Appendices: code snippets	Appendix F	30
Thesis (40%): well motivated, consistent with the data, clearly explained	Section 2.1; the summary of findings at the start of Section 3; Section 4	1
Execution (40%): risk control	Equation (2); cap, capacity and rank weights in Section 3.3 and Appendix C	4
Execution: horizon analysis	Figure 3; HQ12 holding periods in Table 3	7
Execution: robustness checks	Section 3.3, Table 3; placebo, bootstrap, deflated Sharpe, as-known rerun in Appendix B and Appendix C	8
Execution: critical evaluation of the AI’s role	Section 3.4: two explanations tested and not confirmed	8
Originality (20%): novel data use	point-in-time JOLTS and CES vintages by industry (Section 2.2)	2
Originality: prompt design	blinded agents in two teams with a fixed grammar (Figure 2, Table 18); human-audited judge	4
Originality: unique insights	labor flows as slow cost news (Section 3.3); testing explanations of AI behaviour (Section 3.4)	8
AI requirement 1: three substantial interactions	P2, P1 and P4 prompts in Section 2.3; six roles evaluated in Section 3.4	3
AI requirement 2: each documented with prompt, output and critical evaluation	Table 42 and Appendix E	28
AI requirement 3: reflection on strengths and limits	Section 3.4; Section 4	8

The brief names ChatGPT; this term’s course allows Claude, which we used in every AI role (Section 3.4).